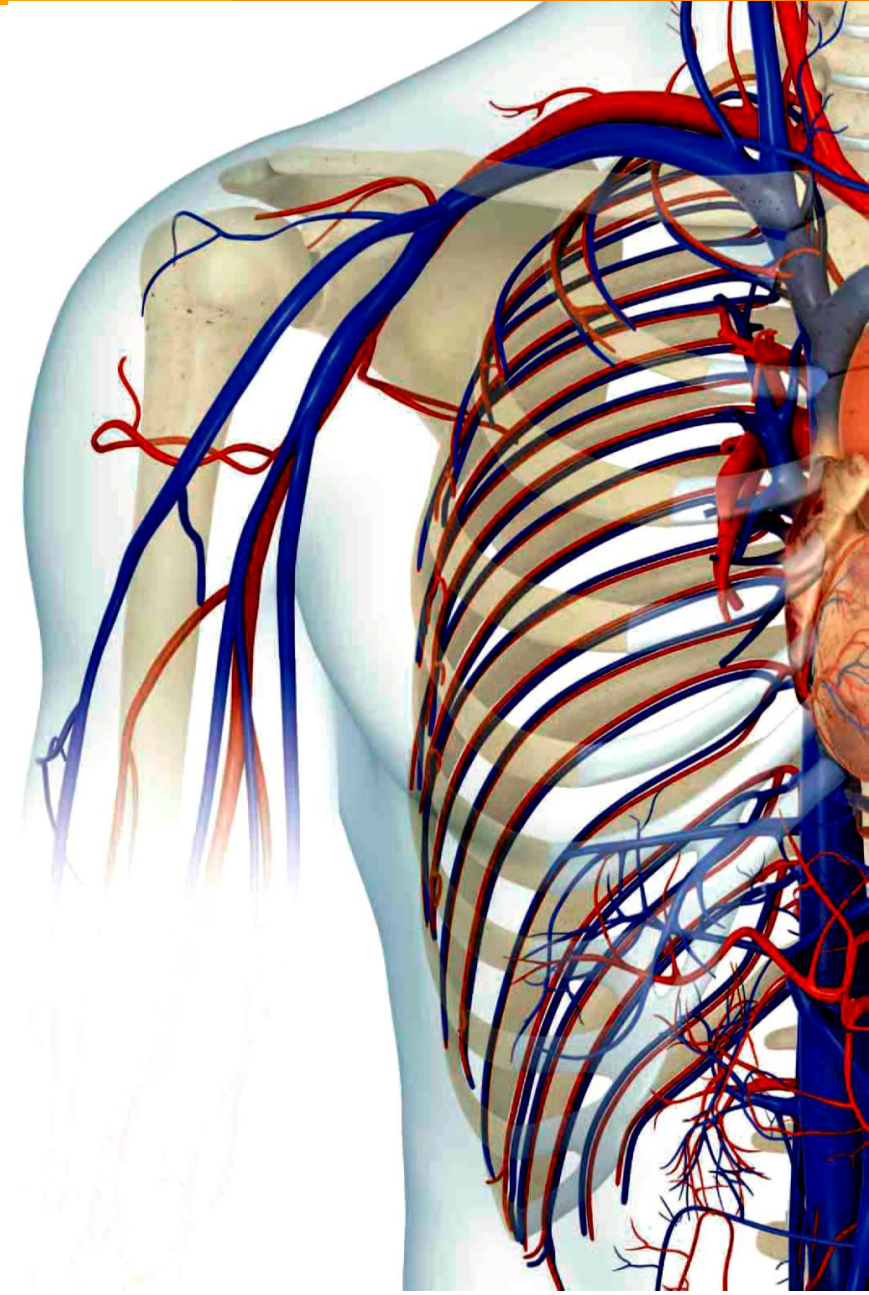


Cardiovascular system-

By: Dr: Nali Abdulkader Maaruf



Cardiovascular system

- Consists of the heart, and blood vessels (arteries, veins & capillaries).
- The heart is located in middle of mediastinum and covered by pericardium.

Blood Vessels

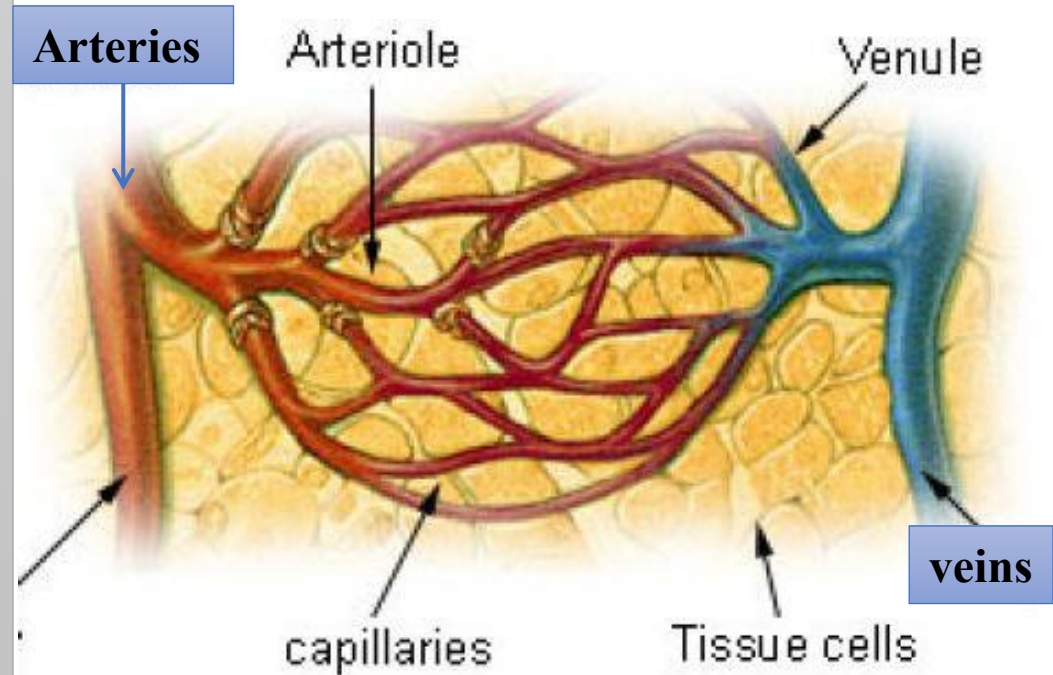
❖ Are of three types, arteries, veins and capillaries.

1. Arteries transport blood from the heart to various tissues of the body by means of its branches.

❖ The smallest arteries are called arterioles.

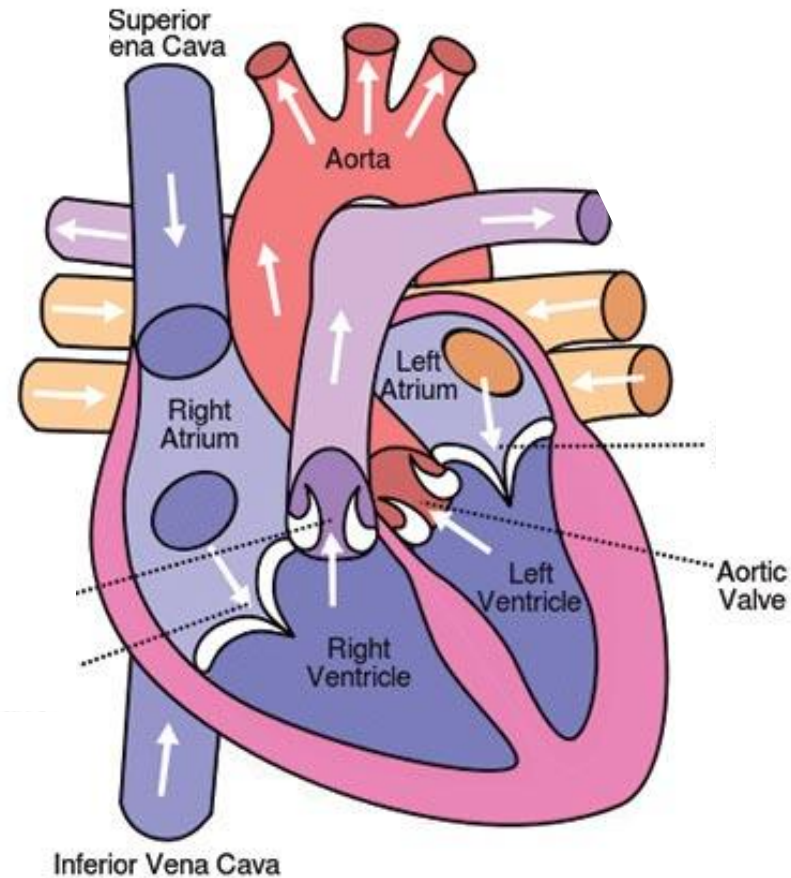
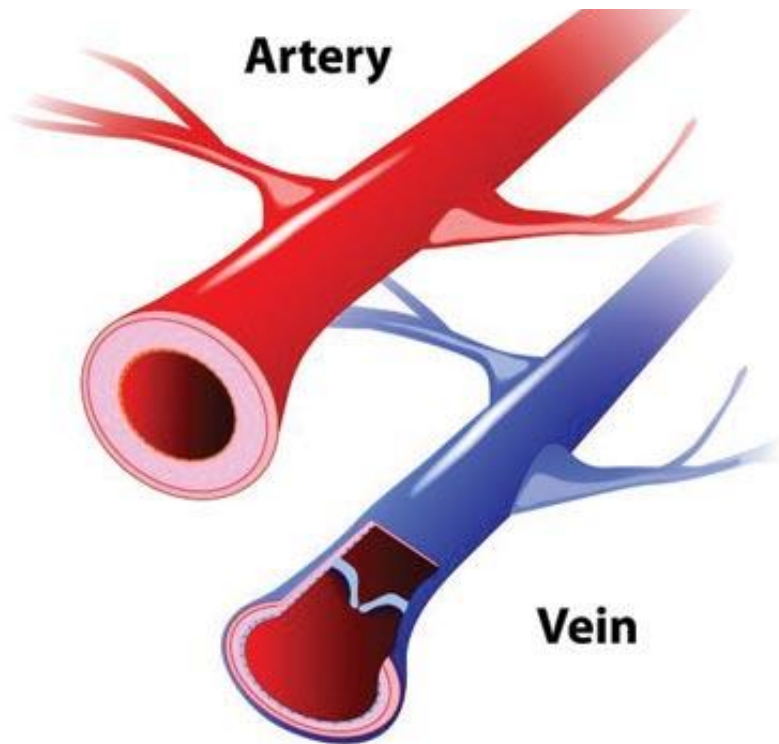
❖ The joining of branches of arteries is called an anastomosis.

❖ Arteries haven't valves.



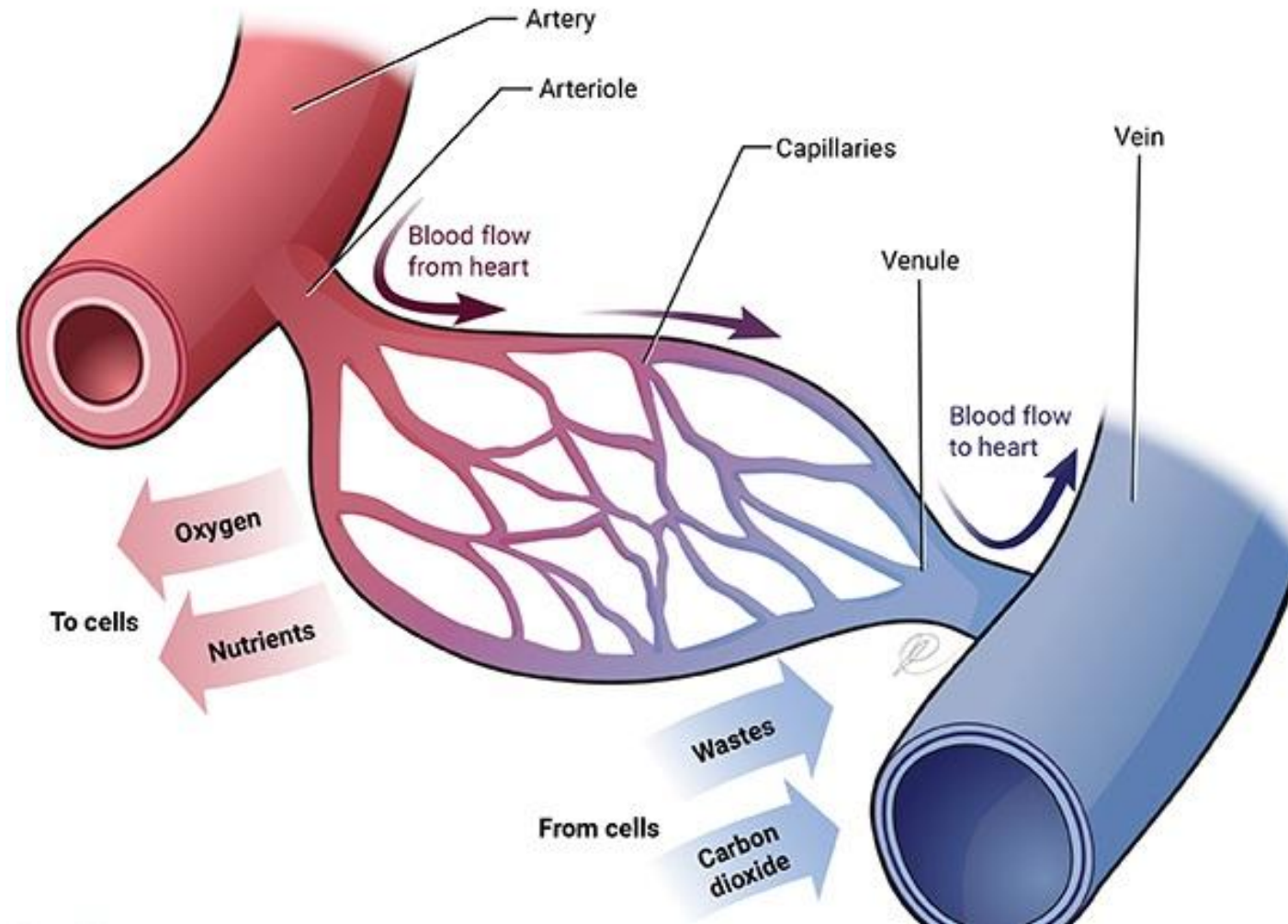
2. Veins

- Are vessels transport blood back to the heart.
- Many of them possess valves.
- Thin wall
- The smallest veins are called venules.

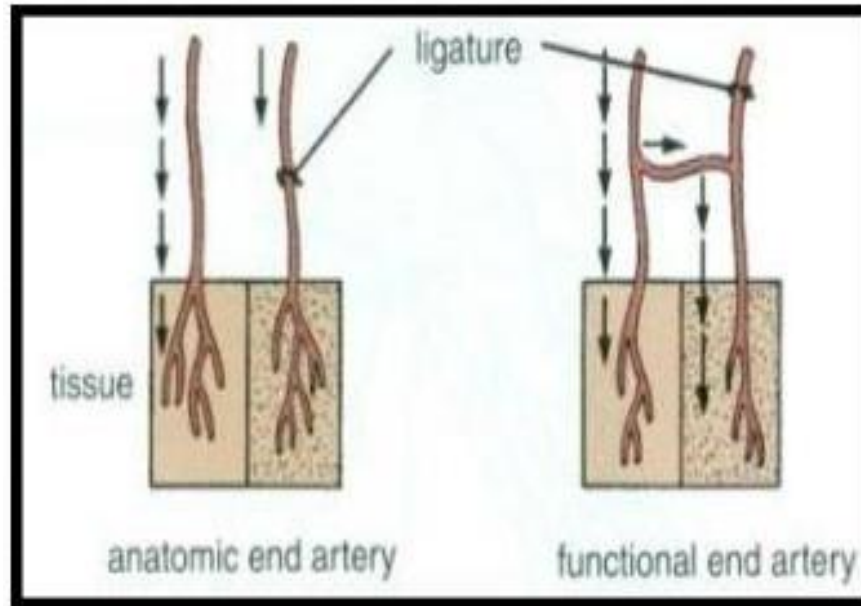


3. Capillaries

- Are microscopic vessels in the form of network connecting the arterioles to venules.



END ARTERIES



Anatomic End artery:

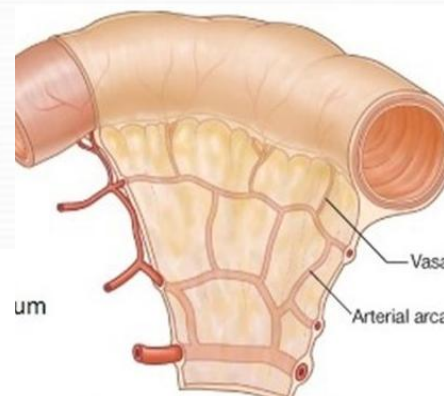
- Its terminal branches **do not anastomose** with the branches of arteries supplying adjacent areas.

Functional End artery:

- Its terminal branches **do anastomose** with those of adjacent arteries but the **anastomosis is insufficient** to keep the tissue alive if one of the arteries is occluded.

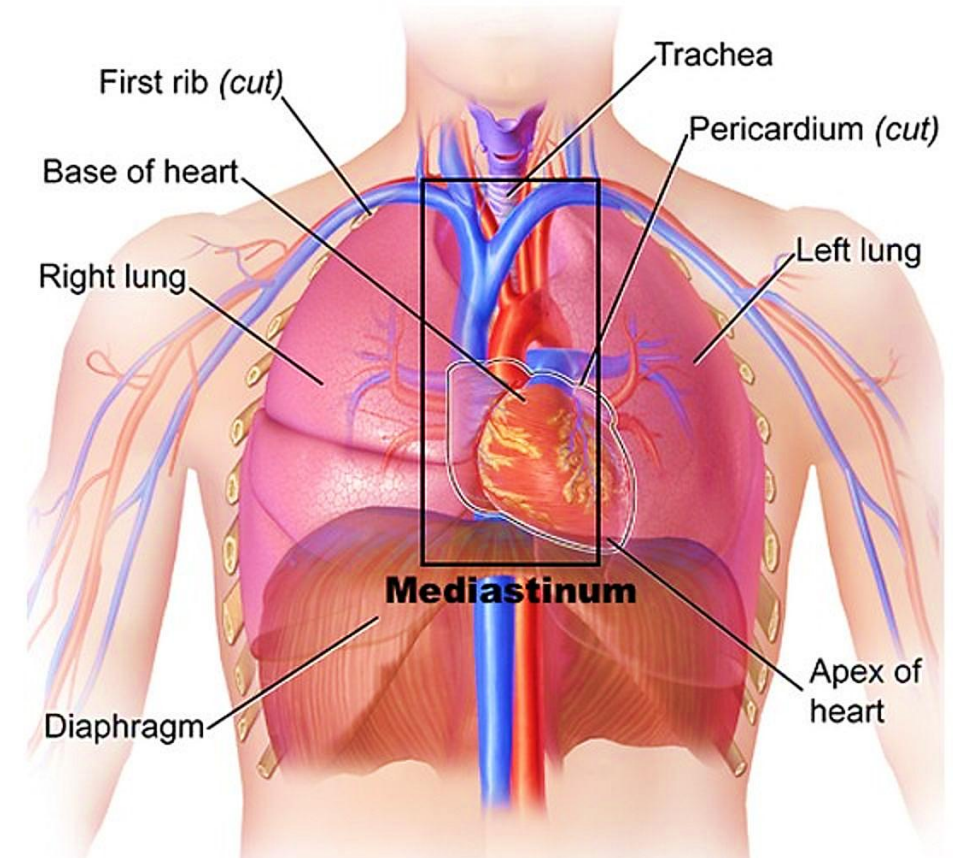
➤ Arterial anastomosis

Union of two ends of arteries ex: arcades of intestine



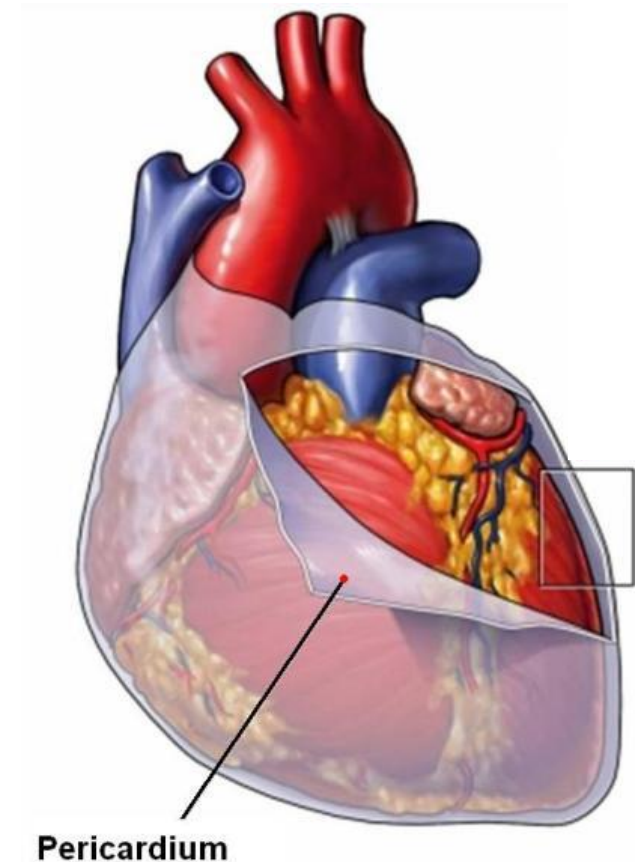
Mediastinum

- The mediastinum is the **central compartment of the thoracic cavity**, located **between the two pleural sacs**.
- It contains most of the **thoracic organs**, and acts as a **conduit for structures traversing the thorax** on their way into the abdomen.



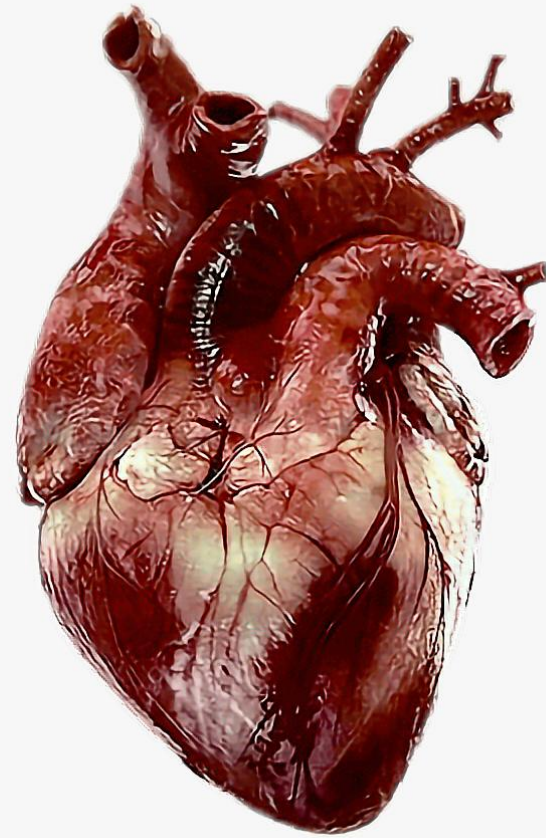
Pericardial layers

- Fibrous pericardium
- Serous pericardium:
 - a. Parietal pericardium.
 - b. Visceral pericardium.



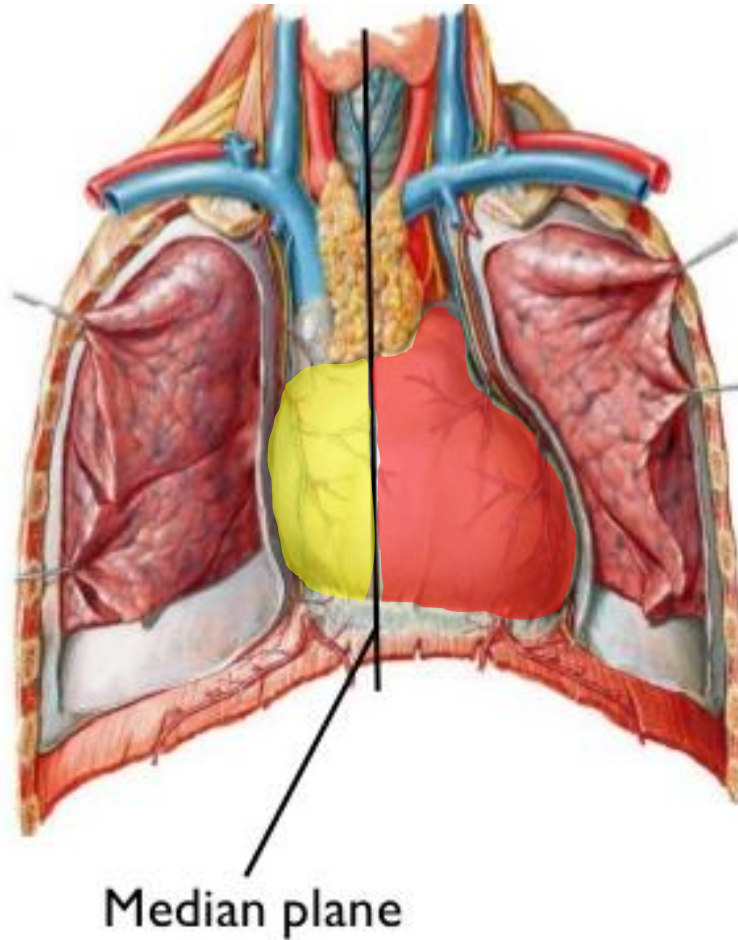
The heart

- **Pyramidal in shape**, pumps and receives blood from all parts of body.
- Covered and **protected by pericardium**.
- Located in the **middle mediastinum** of thoracic cavity between the two lungs.
- Its **1/3rd** is located to **right** of midline, while the other **2/3rd** lies to the **left**.
- Usually has the size equal to the fist of an individual.
- **Has four chambers**; two atria above and two ventricles below.



The heart

One third of the heart
 $\frac{1}{3}^{\text{rd}}$



One third of the heart
 $\frac{2}{3}^{\text{rd}}$

Orientation of heart

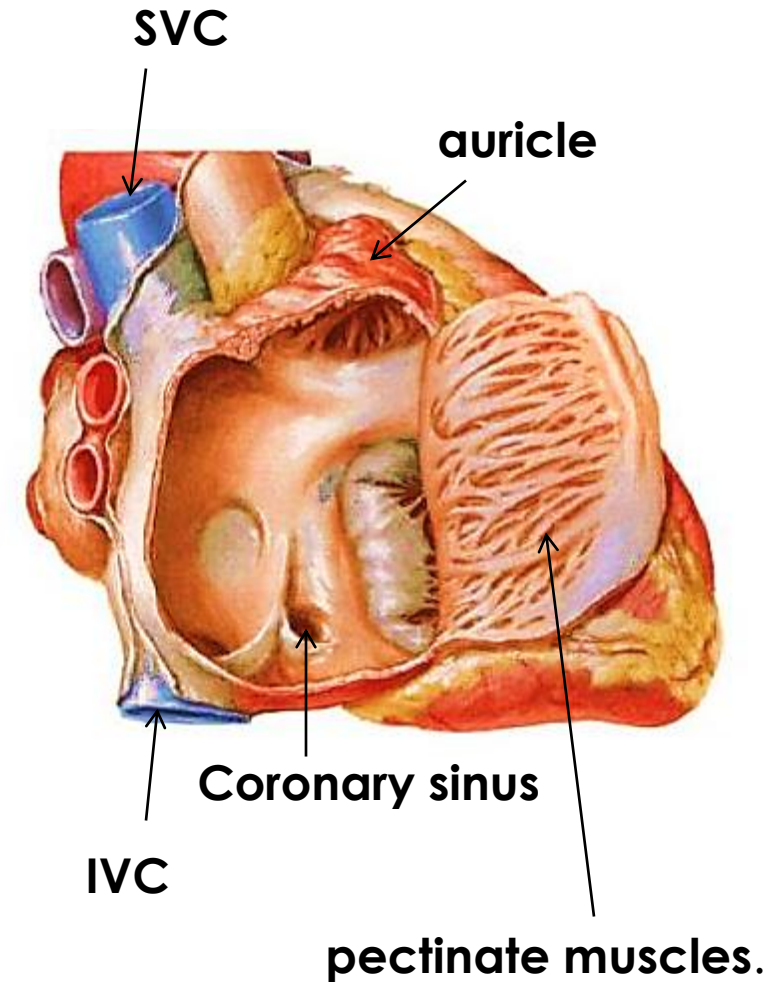
- **Apex:** formed by left ventricle, directed anterior, inferior and to the left. It is located in the **5th left intercostal space 9 cm from the midline.**
- **Base (posterior surface):** formed by the left and right atria, directed posterior, superior to the right. It faces thoracic vertebrae
- **Anterior surface:** lies deep to the sternum and ribs.
- **Inferior surface:** rests on the diaphragm.

Chambers of the Heart

- **Right and left atria** separated by thin interatrial septum.
- **Right and left ventricles** separated by thick muscular interventricular septum.
- Each atrium is continuous with corresponding ventricle by **atrioventricular opening** guarded by **atrioventricular valve** which only **allows the flow of blood from atrium to ventricle**.

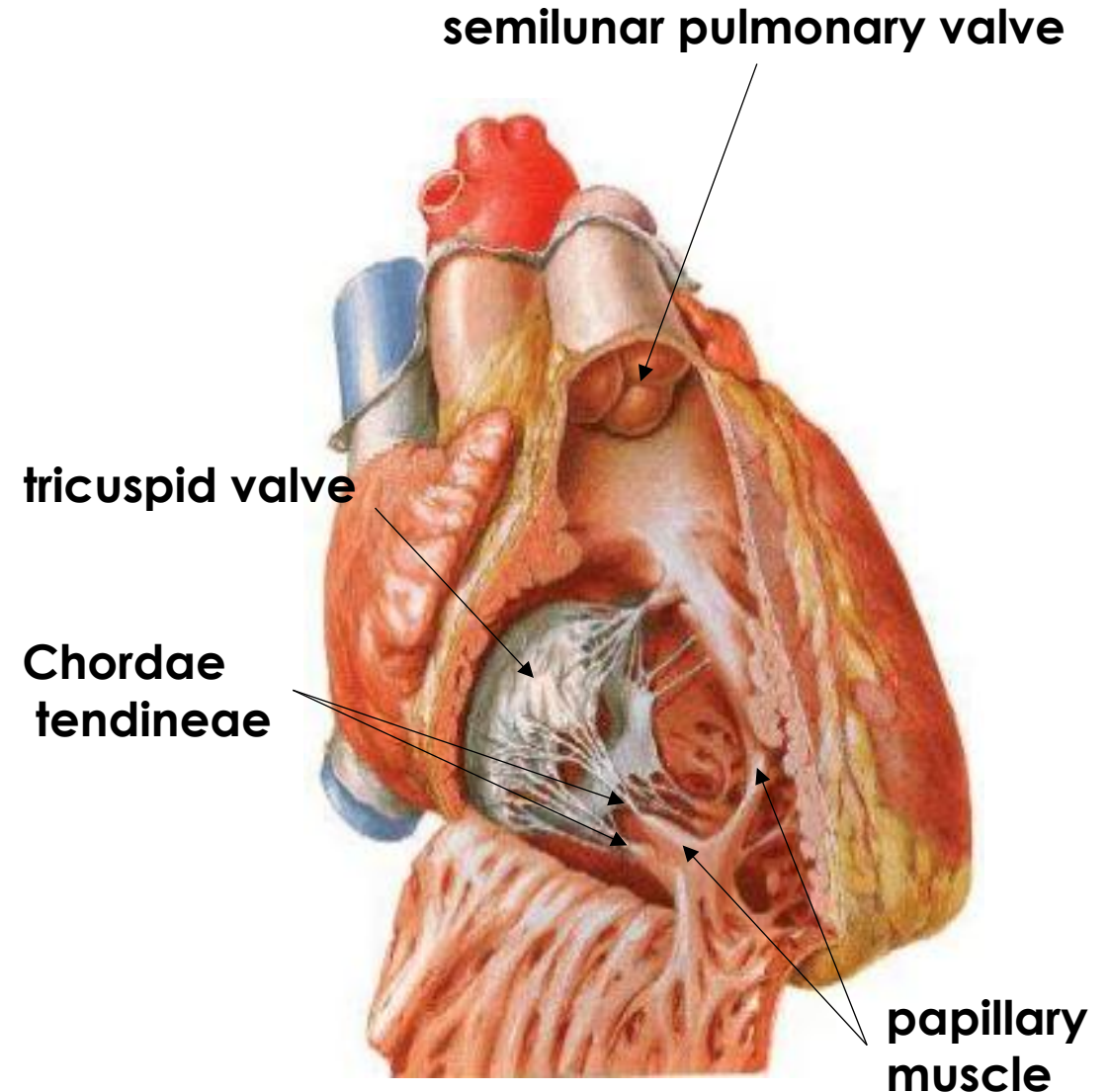
Right atrium

- Has muscular extension called **auricle**.
- The interior of its anterior wall has thin muscular ridges called **pectinate muscles**.
- Has the openings of **three large veins; superior vena cava, inferior vena cava and coronary sinus**.



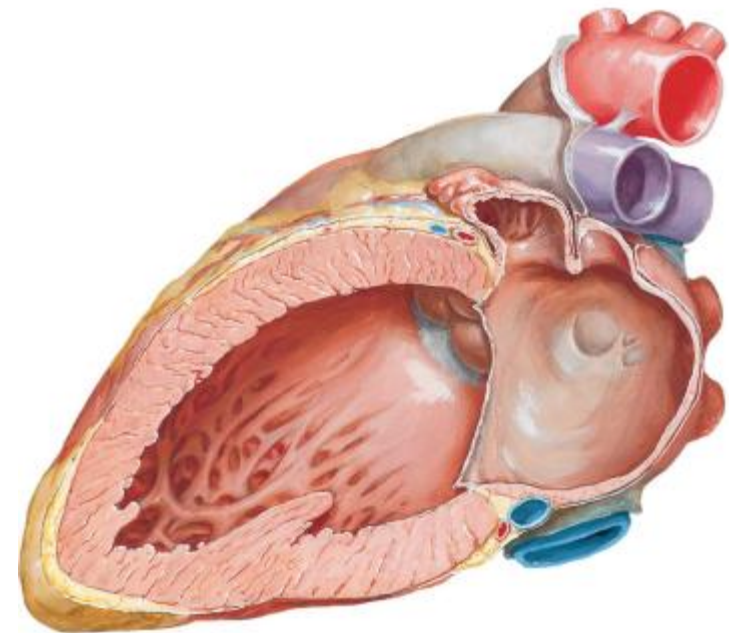
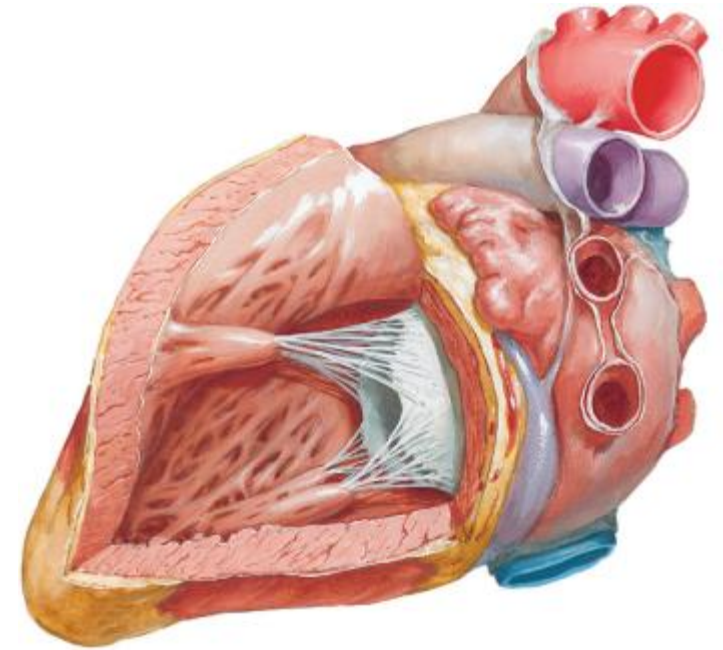
Right ventricle

- Has much **thicker wall than the atria**, and the muscle projects into the cavity called **papillary muscle** and ends as ridges called **Chordae tendineae**.
- Receives **deoxygenated blood from right atrium** via right atrioventricular orifice.



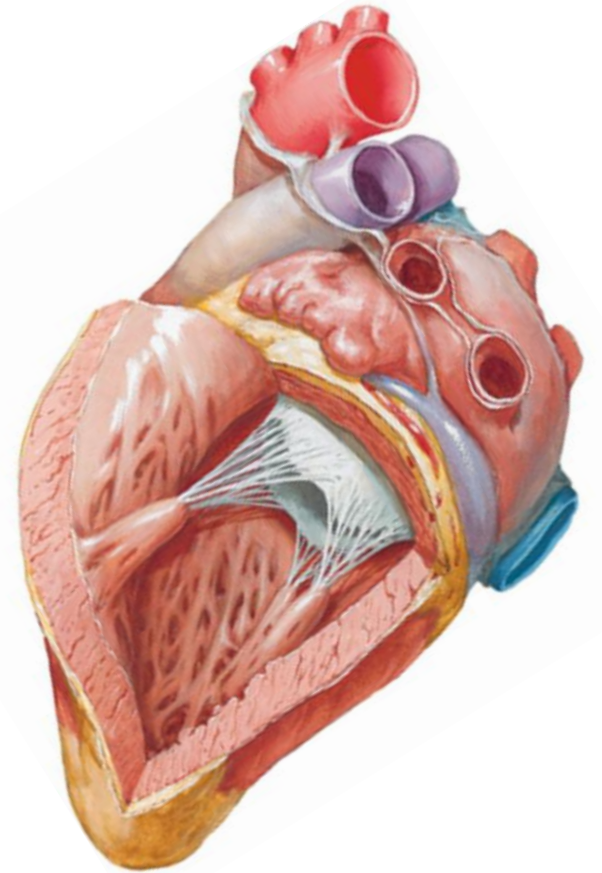
Left atrium

- Has muscular extension called **auricle**.
- Receives **oxygenated blood** from the lungs by **four pulmonary veins** (2 right and 2 left).
- Pumps blood into **left ventricle** through left **atrioventricular** or **bicuspid** or **mitral valve**.

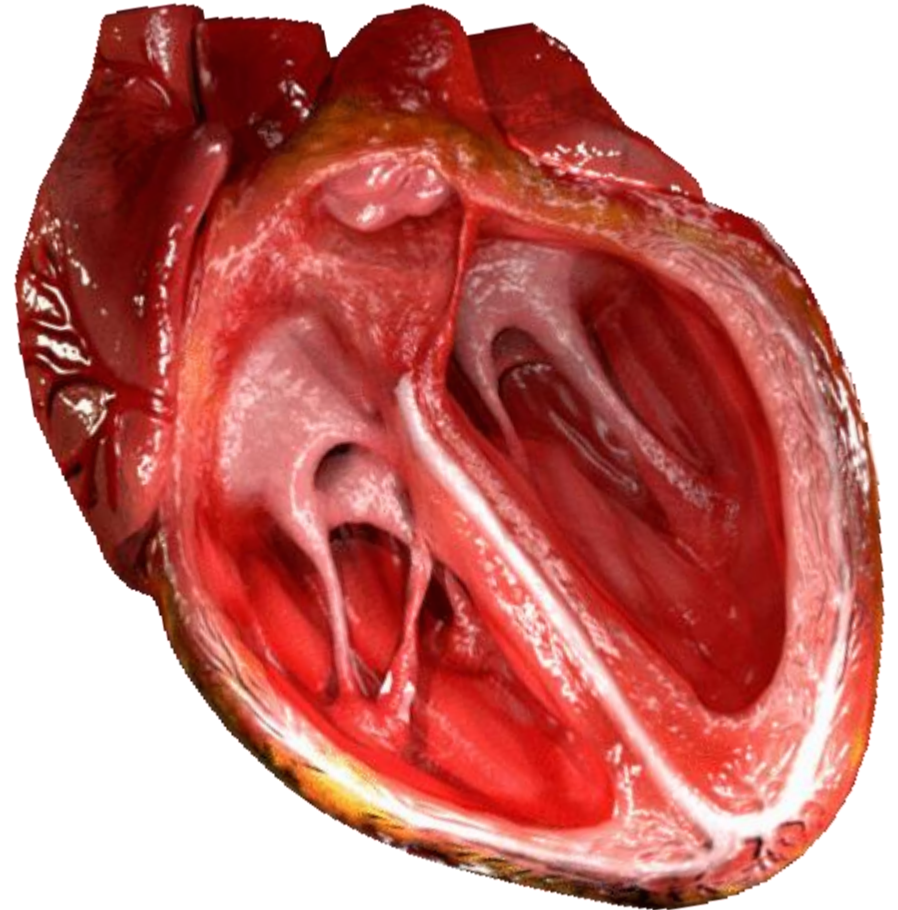
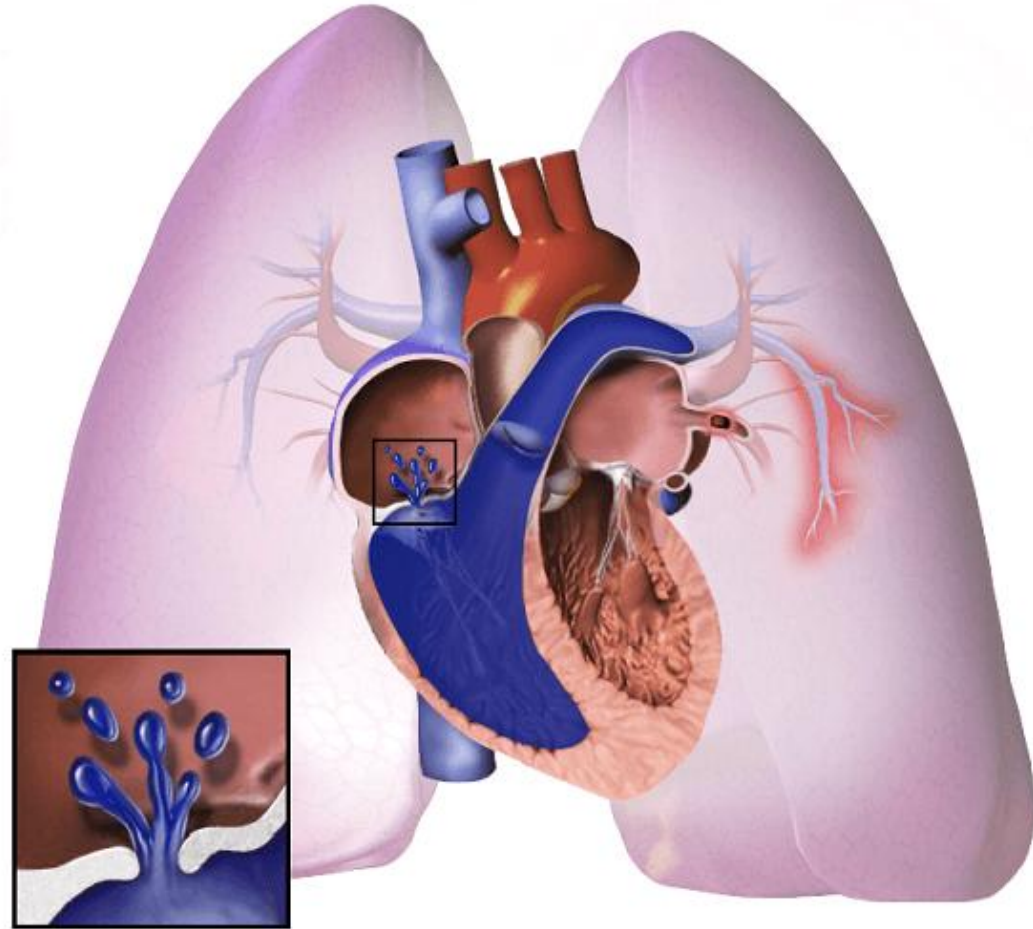


Left ventricle

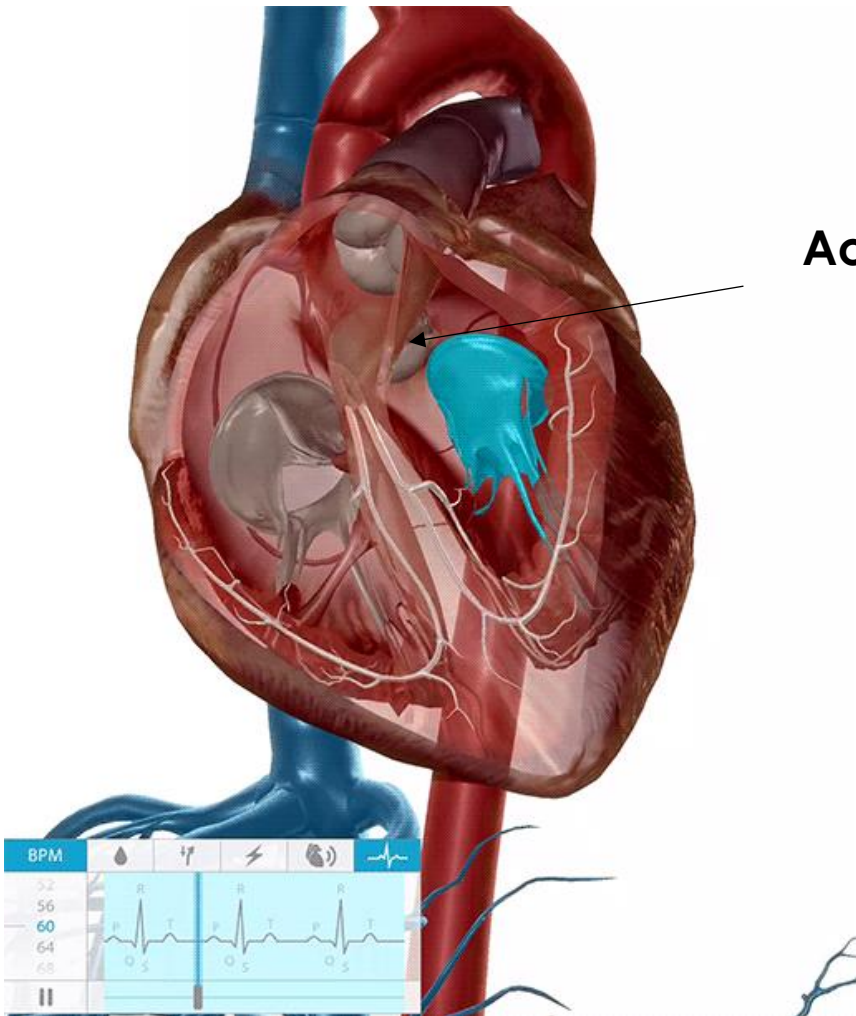
- Forms the **apex of heart**.
- **Its wall is three times thicker than that of right ventricle**, and the muscle projects into the cavity as ridges called **papillary muscle**.
- Receives **oxygenated blood** from left atrium via **bicuspid valve** and **ejects it to the aorta** by through **semilunar aortic valve**.



Chambers of the heart

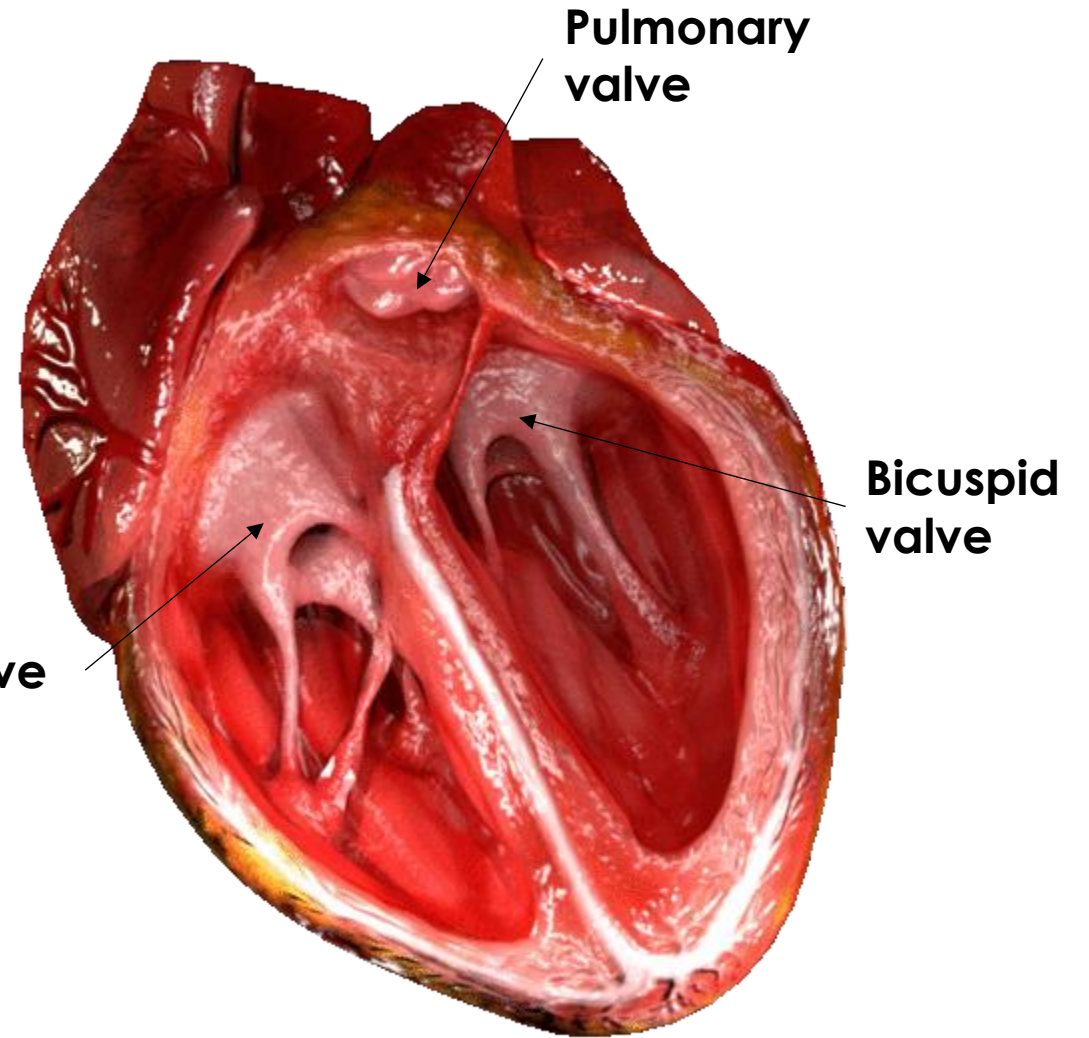


Valves of the heart



Aortic valve

Tricuspid valve



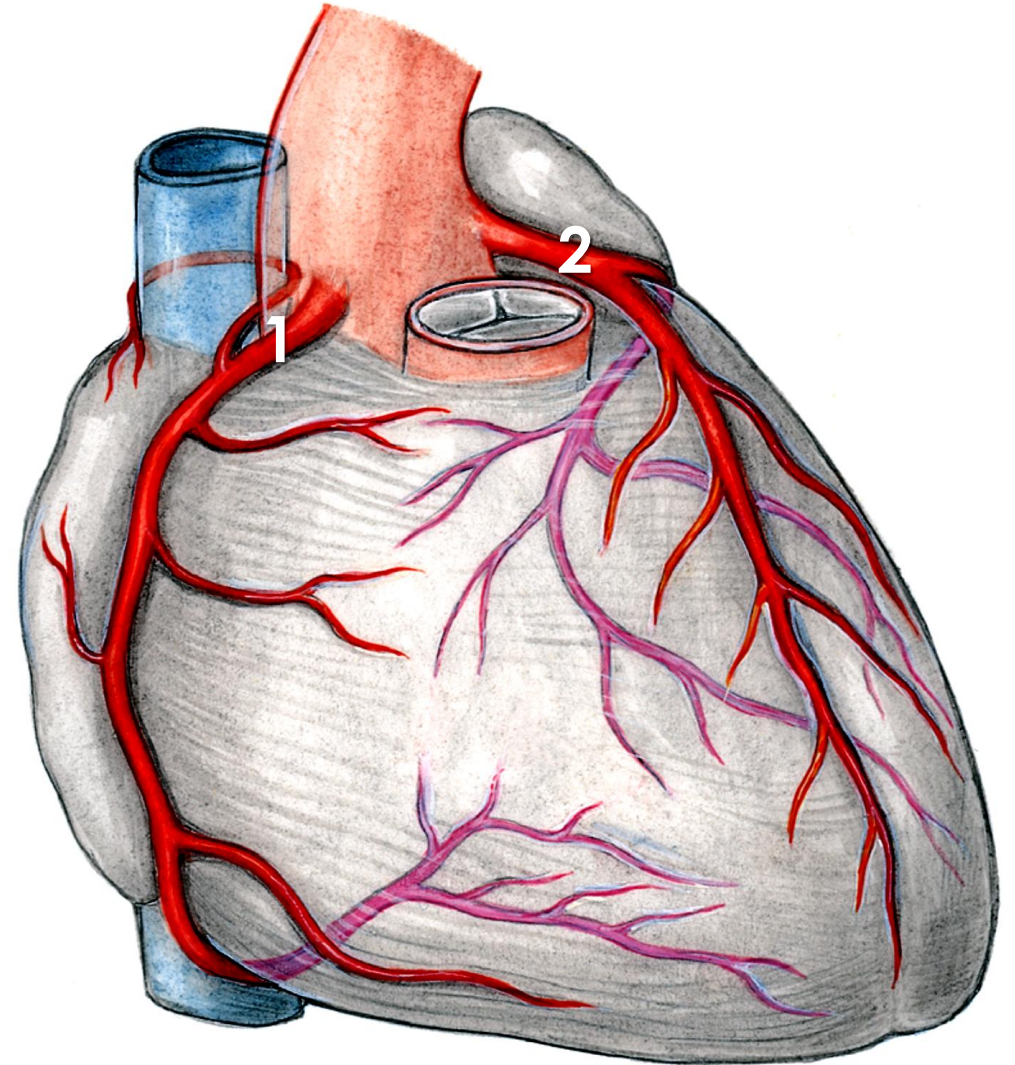
Pulmonary valve

Bicuspid valve

Blood supply of the heart

Arterial supply:

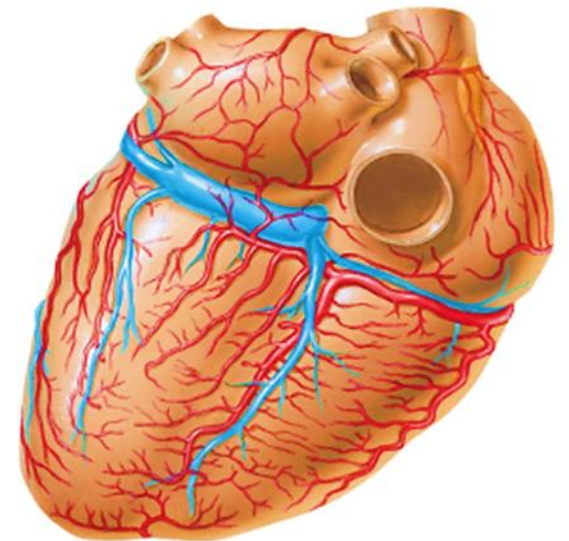
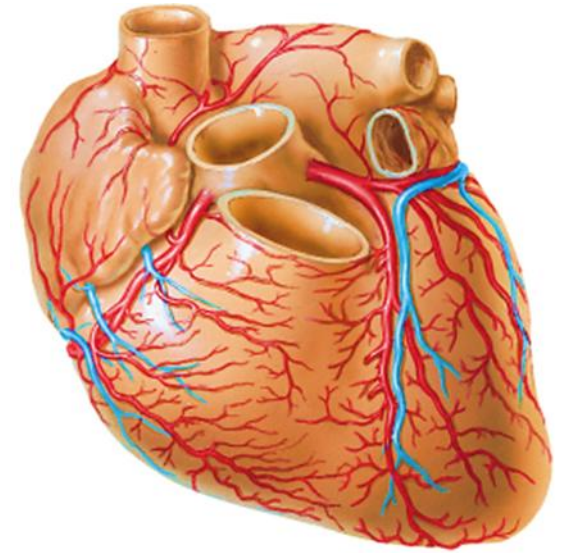
- Branches of ascending aorta
 1. The left coronary artery
 2. The right coronary artery.



Blood supply of the heart

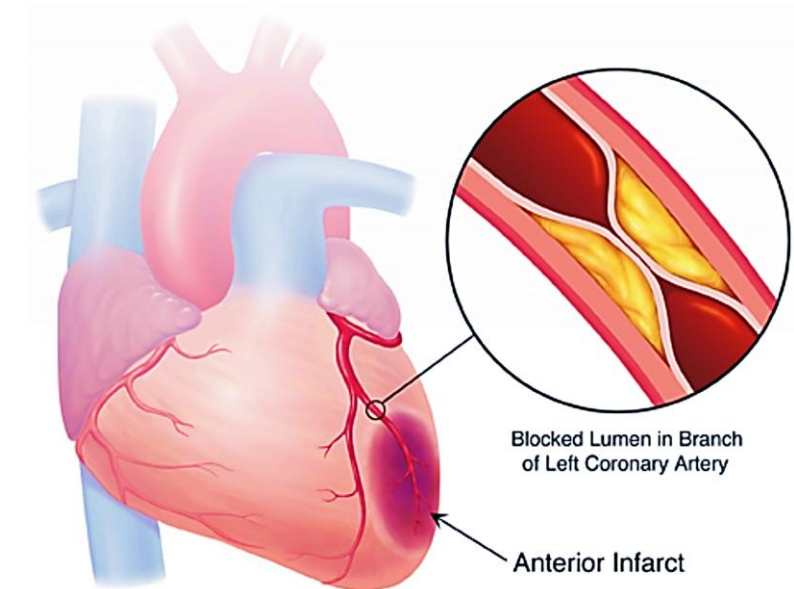
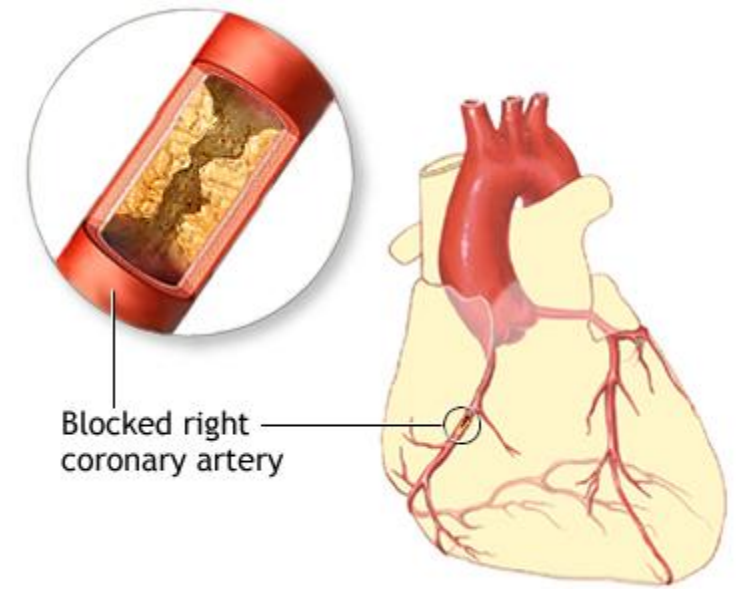
Cardiac veins drainage:

- Collects blood from the wall of heart into **coronary sinus**.
- The coronary sinus **opens into the right atrium**.



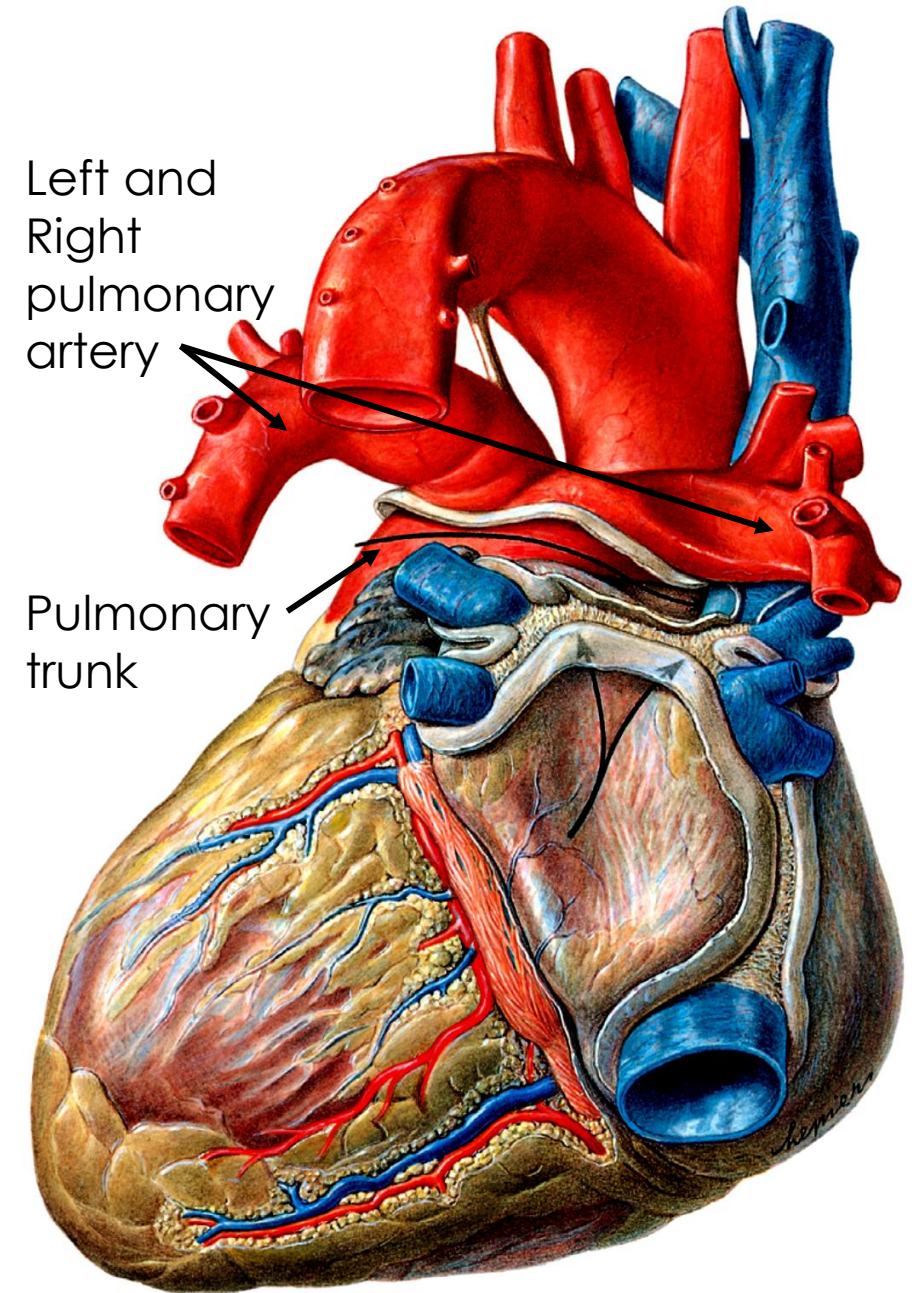
Clinical notes

- Their narrowing results in **angina pectoris** which as a result of **decrease in oxygen** supply of cardiac muscles.
- Their **complete obstruction** results in **myocardial infarction (MI)** due to **death of cardiac muscles**.



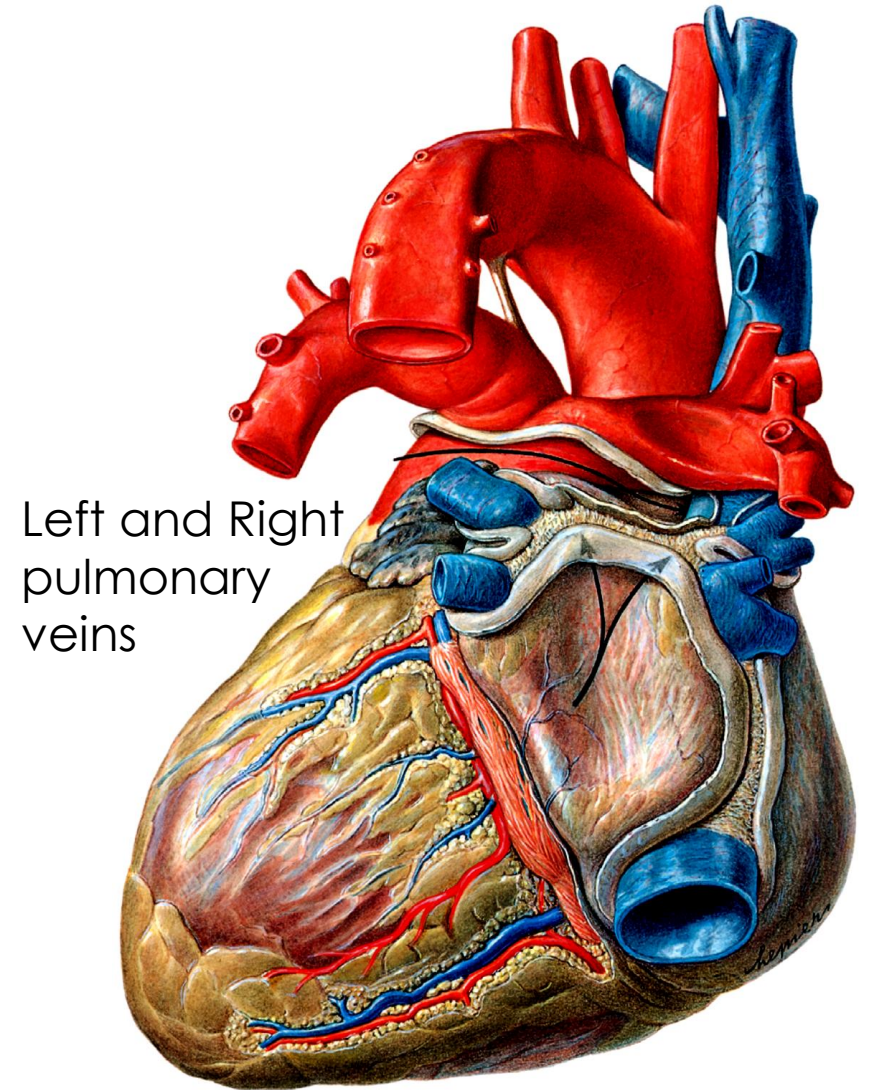
Pulmonary trunk

- The **pulmonary arteries** receive **deoxygenated blood** from the right ventricle, and **deliver it to the lungs**.
- The pulmonary trunk then splits into **the right and left pulmonary arteries** each for one lung.



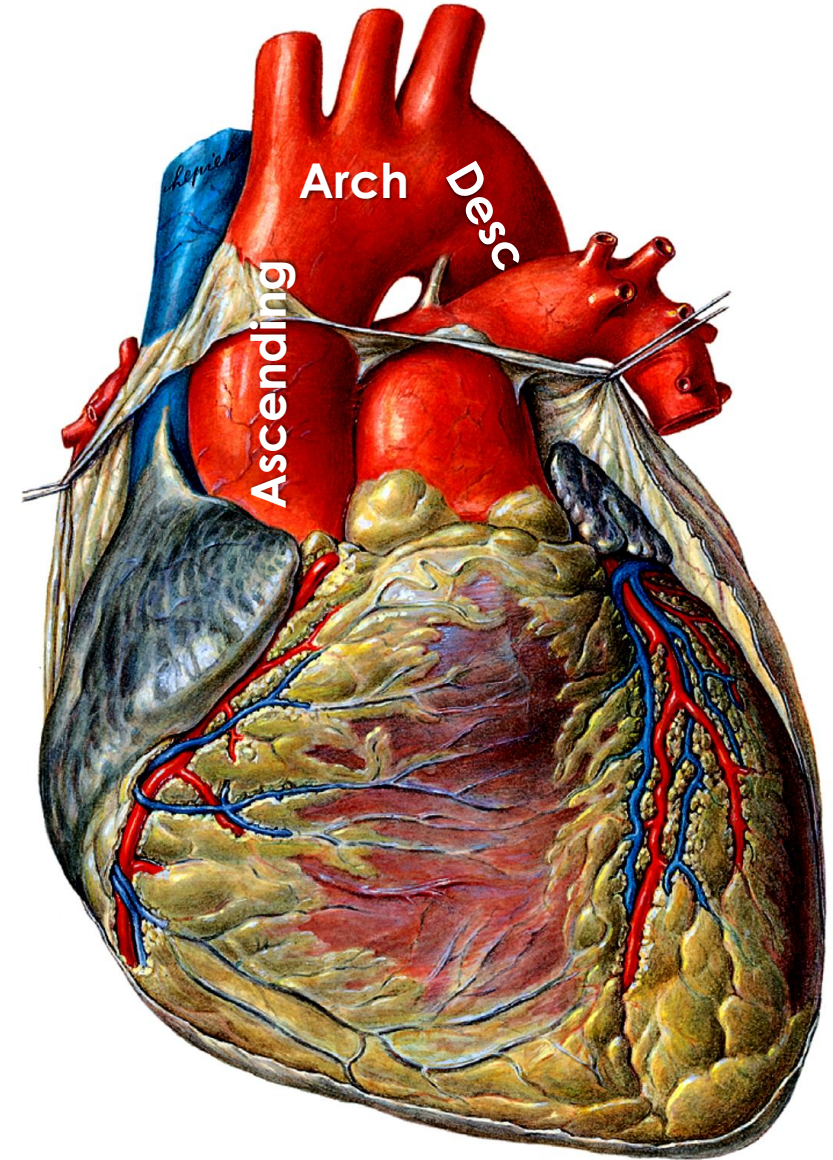
Pulmonary veins

- The pulmonary veins receive **oxygenated blood** from the lungs, and deliver it to the left atrium.
- There are **four pulmonary veins**, two for each lung



Aorta

- The aorta receives **oxygenated** blood from the left atrium, and **delivers it to the body**.
- It **begins at the left ventricle**, from which it is separated by the **aortic valve**.
- It then curves laterally, towards the left, forming the **arch of aorta**.

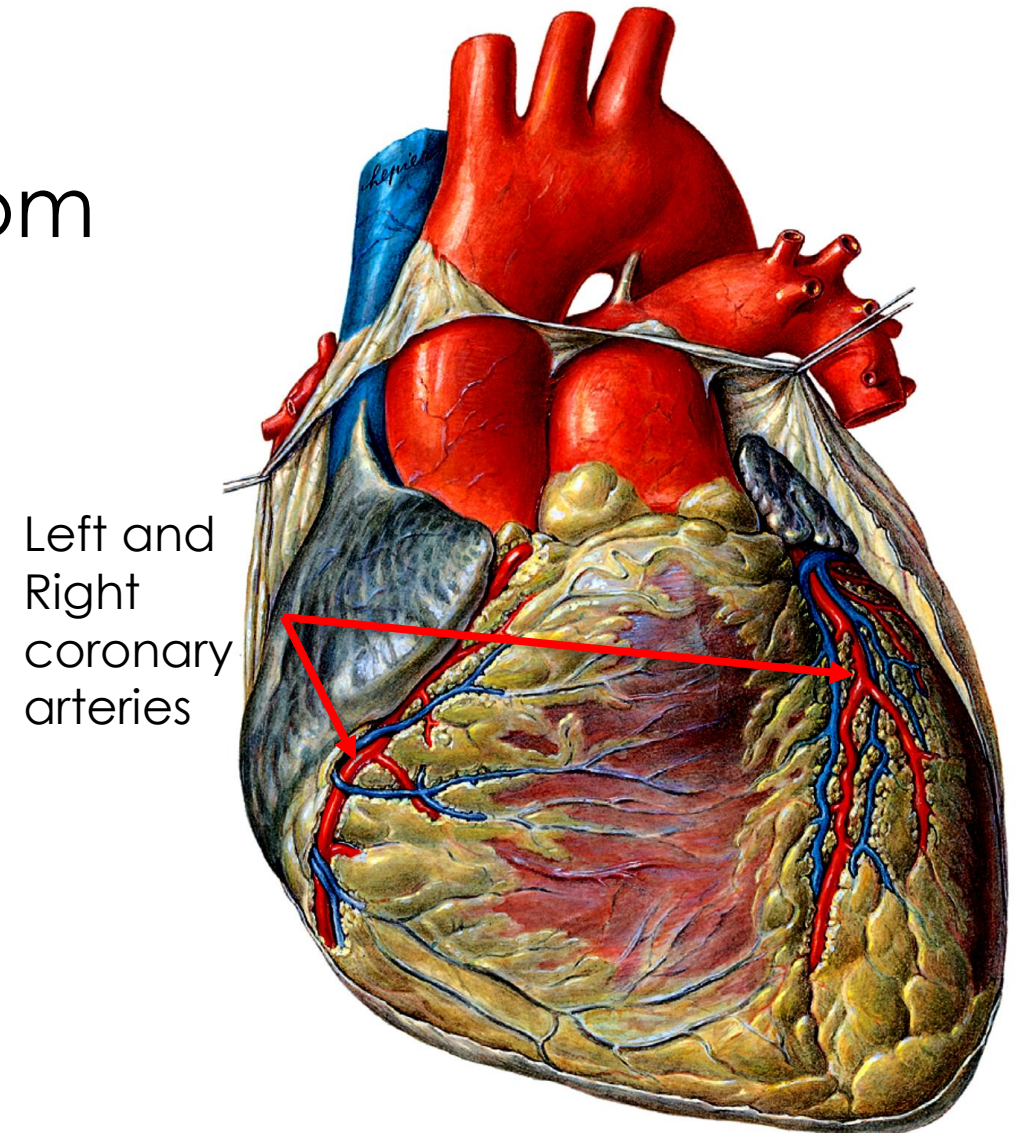


Ascending aorta

- The ascending aorta arises from the **aortic orifice** from the left ventricle and ascends to become the aortic arch.

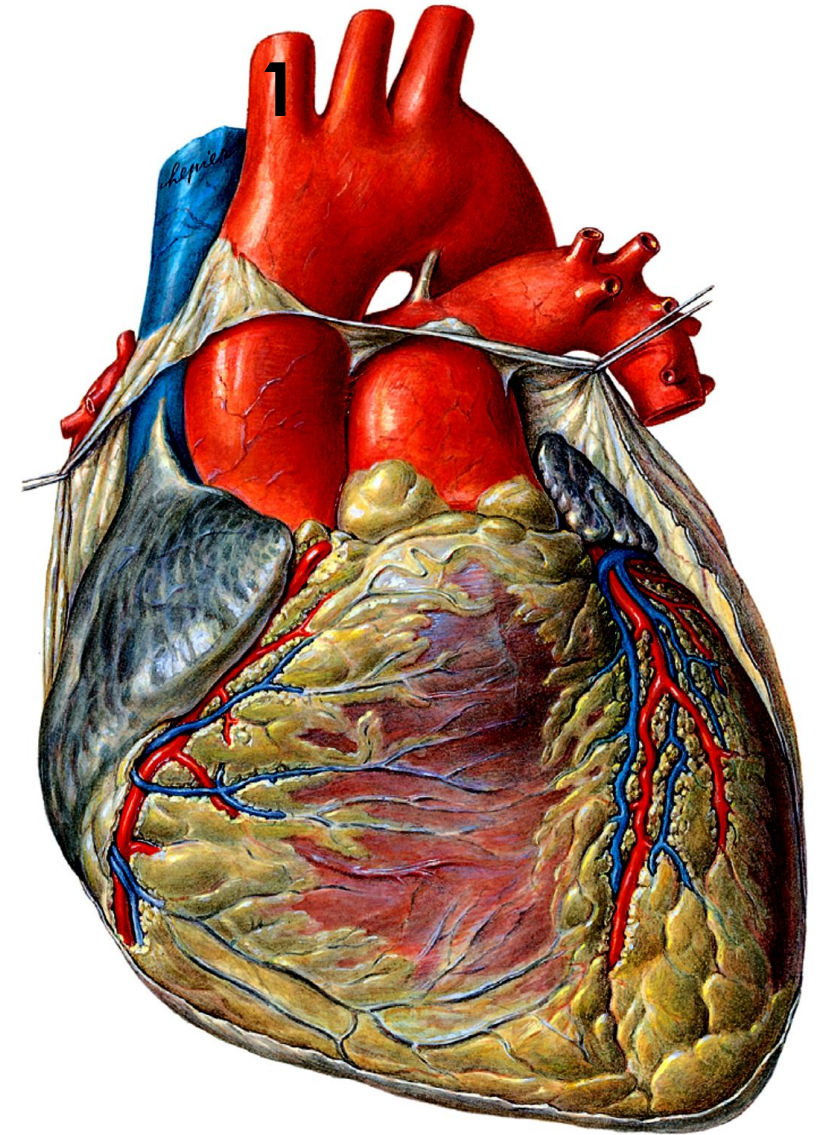
Branches

- The **left and right coronary arteries** that supply the myocardium.



Arch of the aorta

- The arch of the aorta gives rise to three major arteries:
1. **Brachiocephalic artery** – supplies blood to the **brain, head, and the right side of the thorax.**



Arch of the aorta

- The arch of the aorta gives rise to three major arteries:

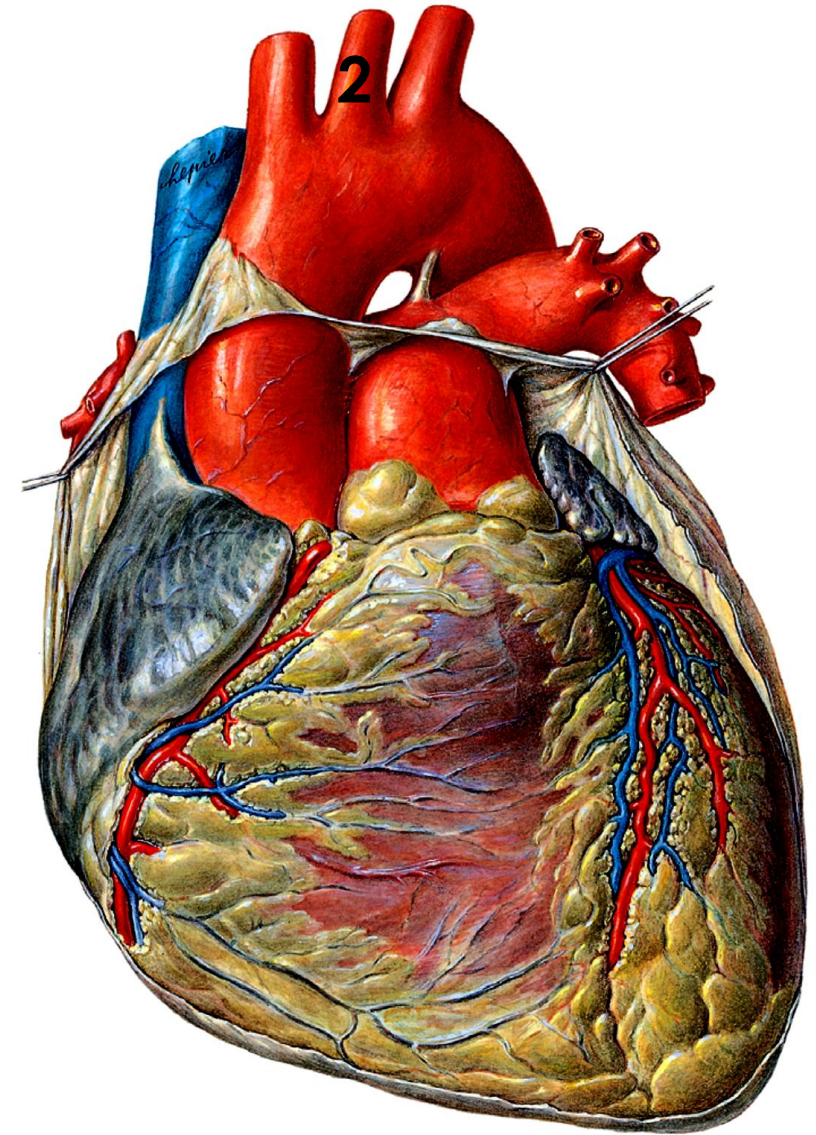
1. Brachiocephalic artery – divided into:

- a. Right common carotid artery.
- b. Right subclavian artery.



Arch of the aorta

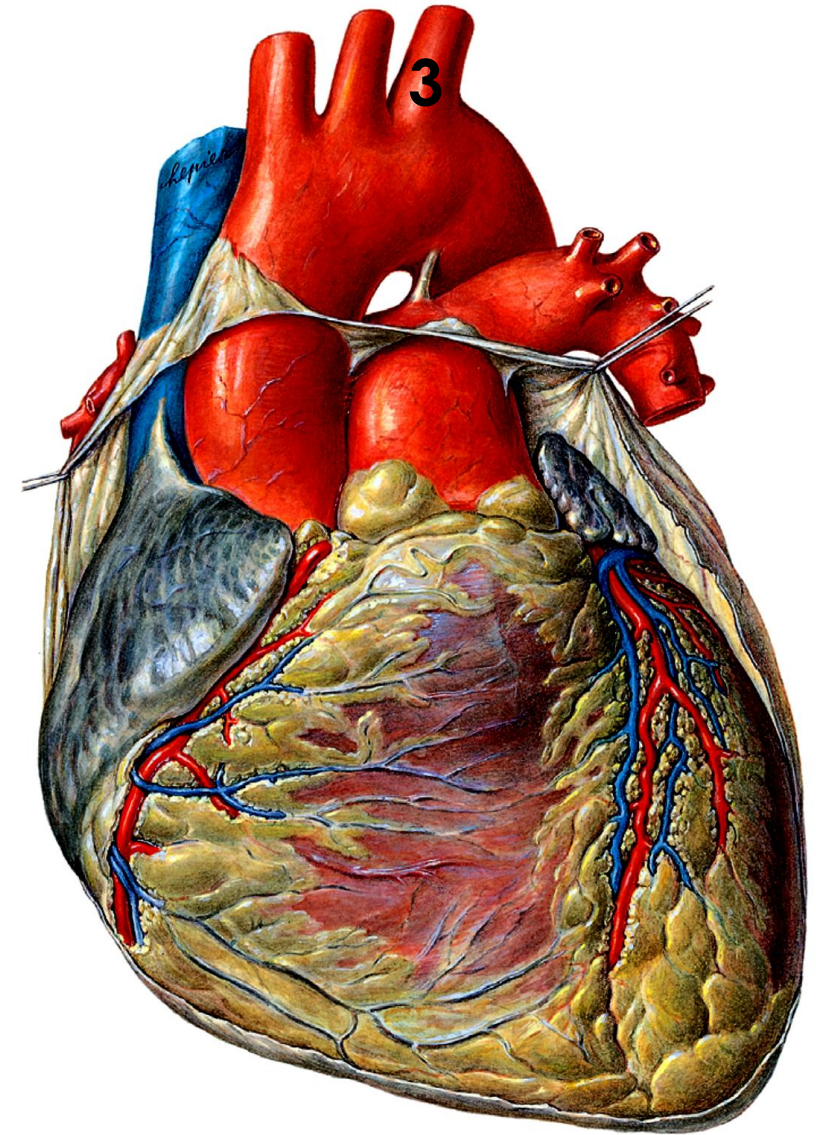
2. **Left common carotid artery** – supplies blood to the **left side of the neck and head.**



Arch of the aorta

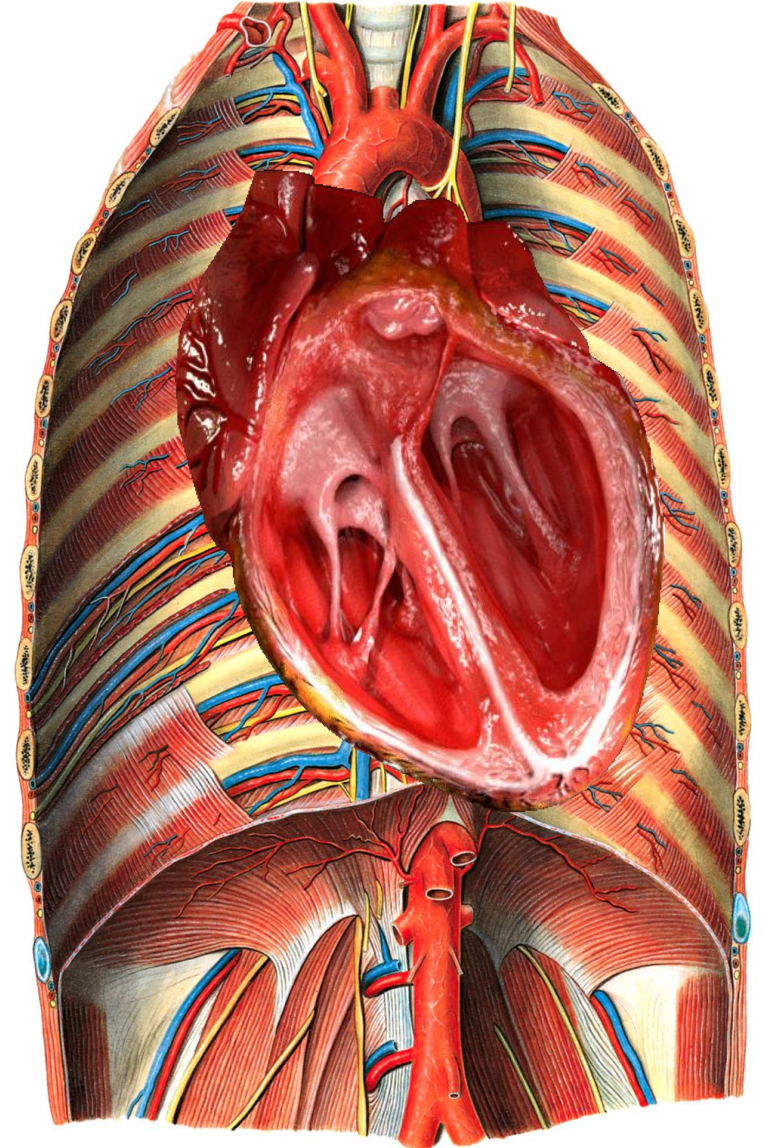
3. Left subclavian artery –
supplies blood to the left
side of the thorax.

After the arch of the aorta, the aorta
then becomes the **descending aorta**
becoming the **thoracic and abdominal**
aorta.



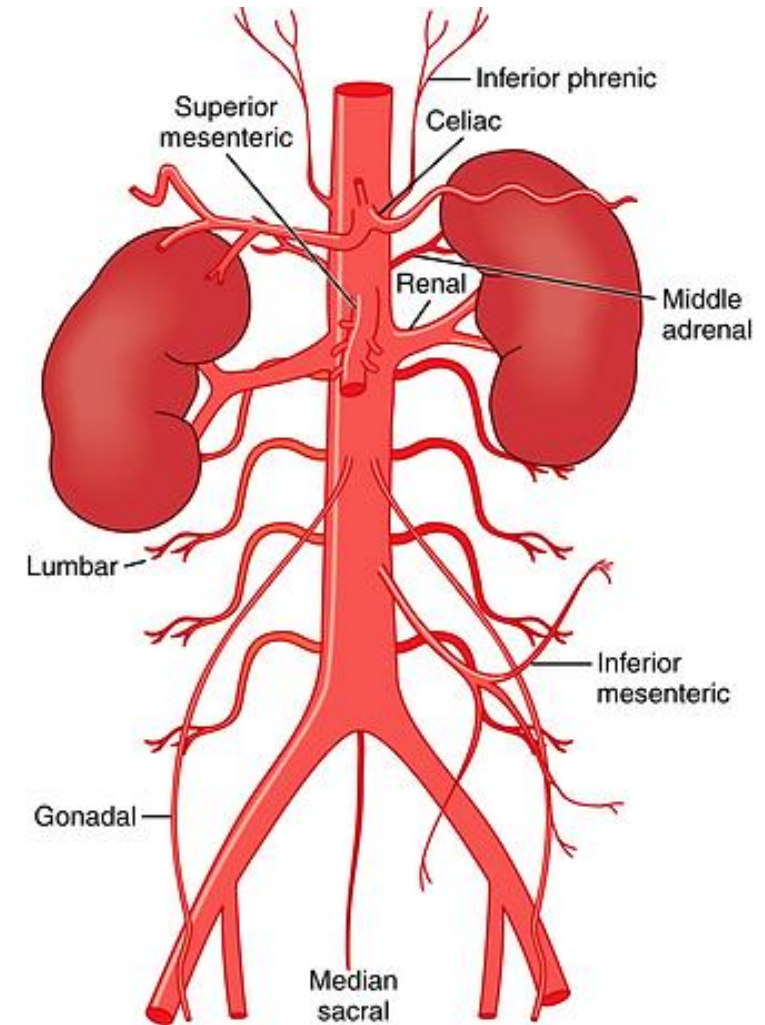
Thoracic aorta (descending)

- The thoracic aorta extends from the level of **T4 to T12**, continuing from the aortic arch.
- It leaves the thorax via the **aortic hiatus** in the diaphragm, and becomes the abdominal aorta.



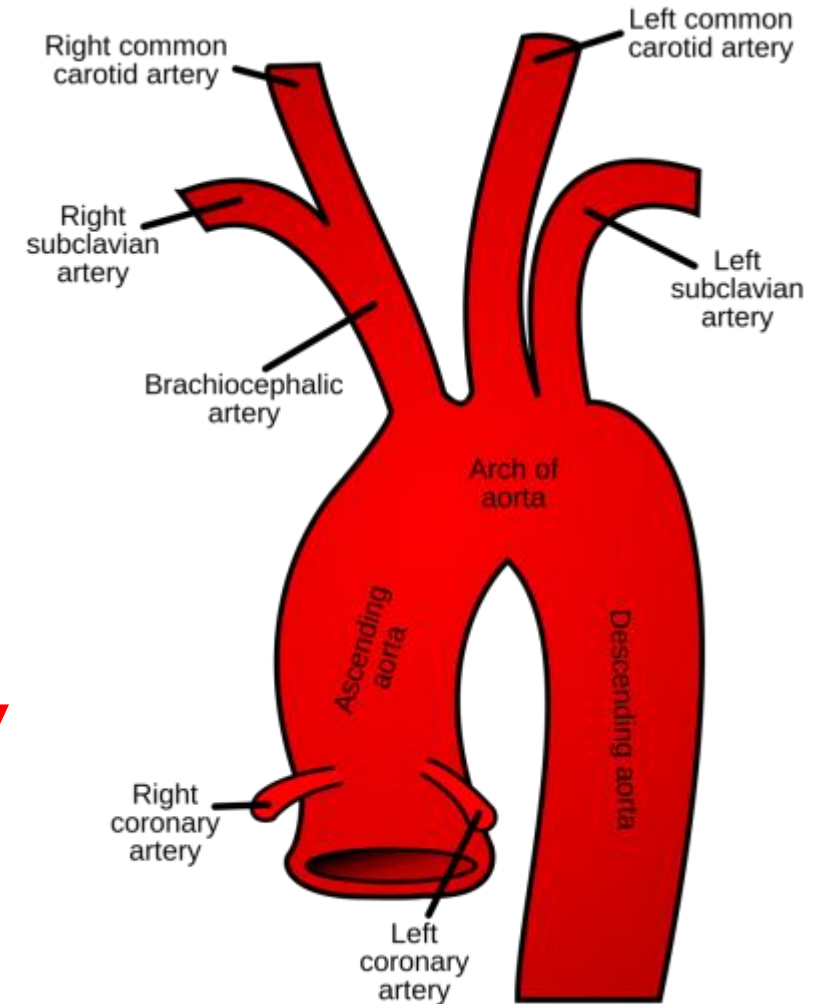
Abdominal aorta

- The abdominal aorta is a **continuation of the thoracic aorta** beginning **at the level of the T12 vertebrae**.
- It is approximately 13cm long and **ends at the level of the L4 vertebra**.



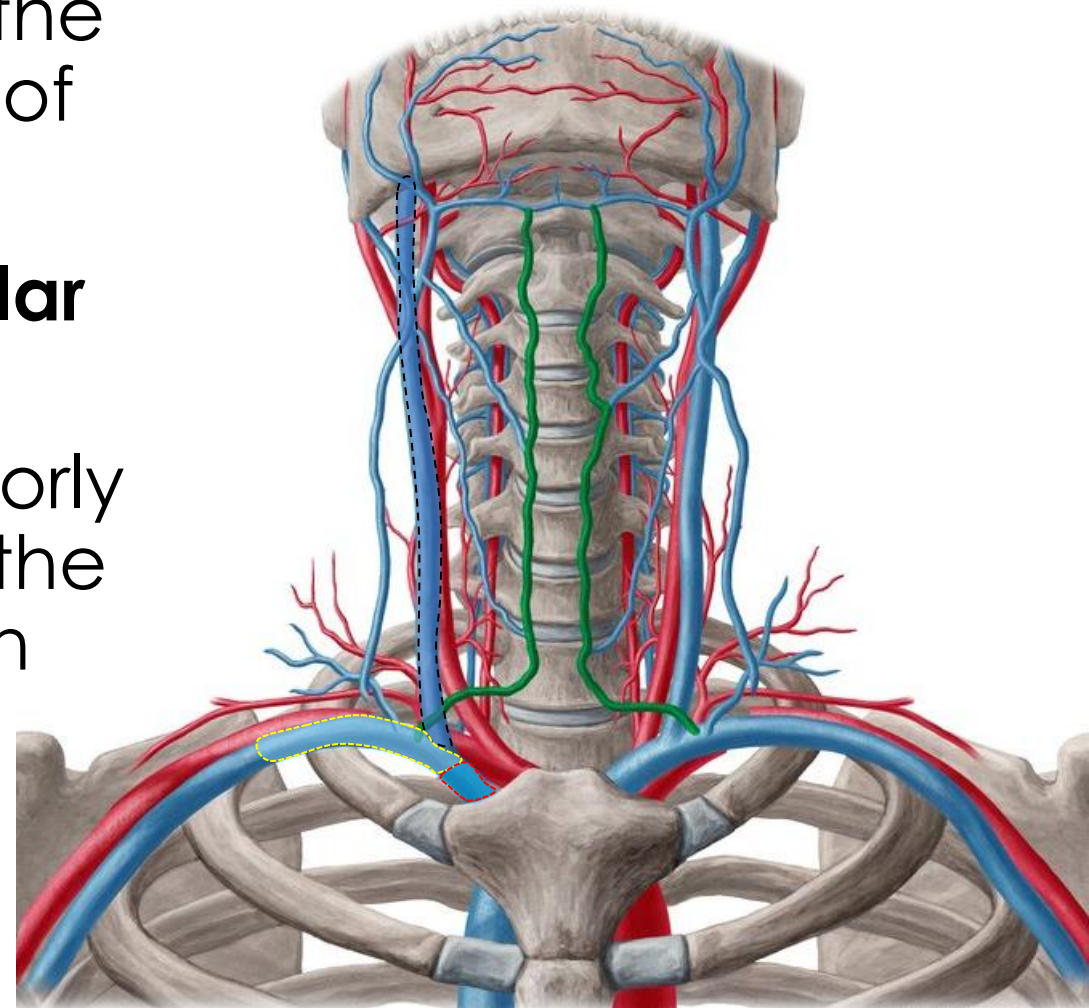
Common carotid artery

- The head and neck receives its blood supply through the **carotid and vertebral arteries**.
 1. The **right common carotid artery** arises from the brachiocephalic trunk.
 2. The **left common carotid artery** branches directly from the arch of aorta.

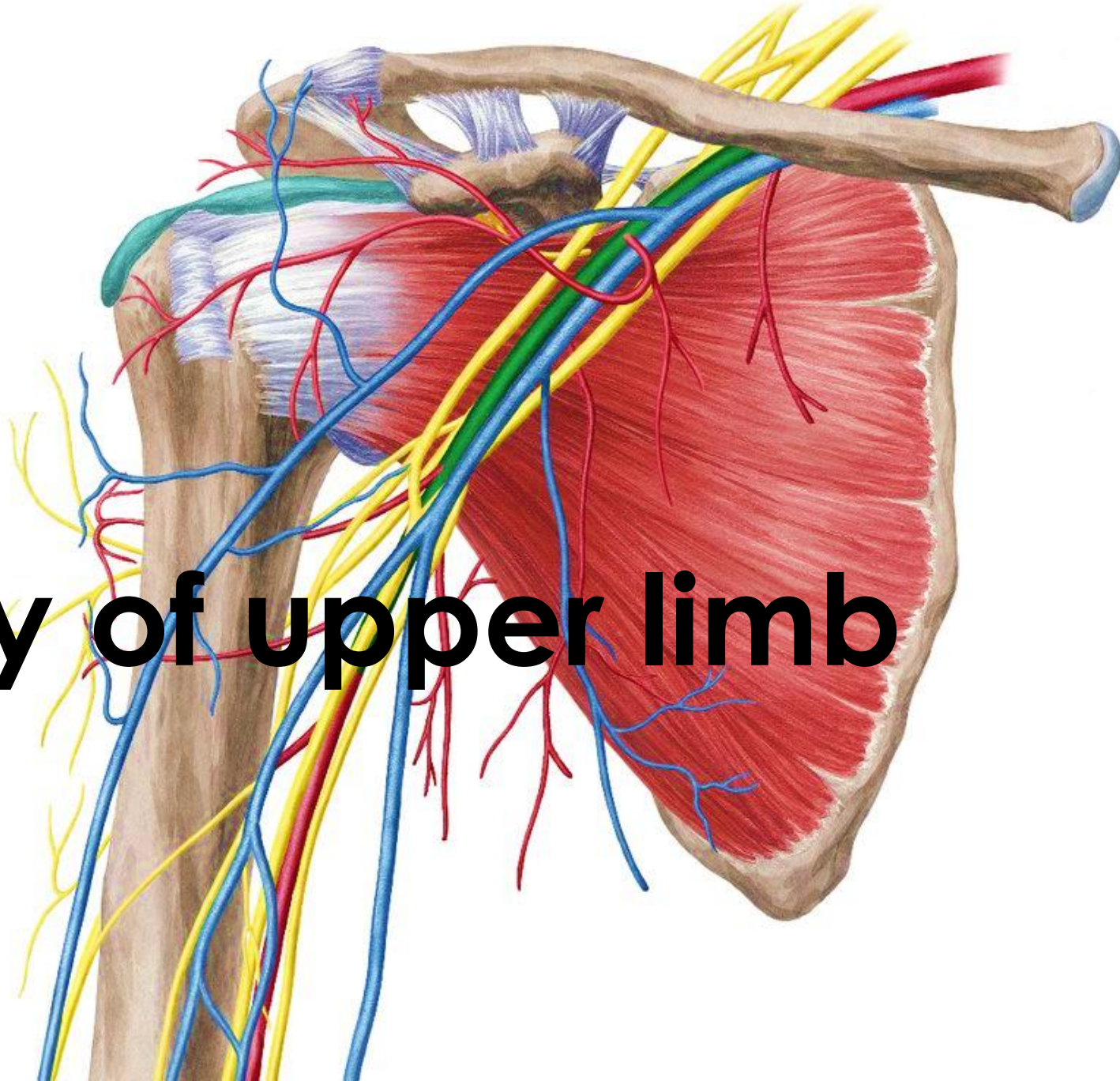


Internal jugular vein

- The internal jugular vein begins in the cranial cavity, as an continuation of the **sigmoid sinus**.
- The vein exits the skull via the **jugular foramen**.
- At the bottom of the neck, posteriorly to the sternal end of the clavicle, the **internal jugular vein** combines with the **subclavian vein** to form the **brachiocephalic vein**.



Blood supply of upper limb



Arteries of the upper limb

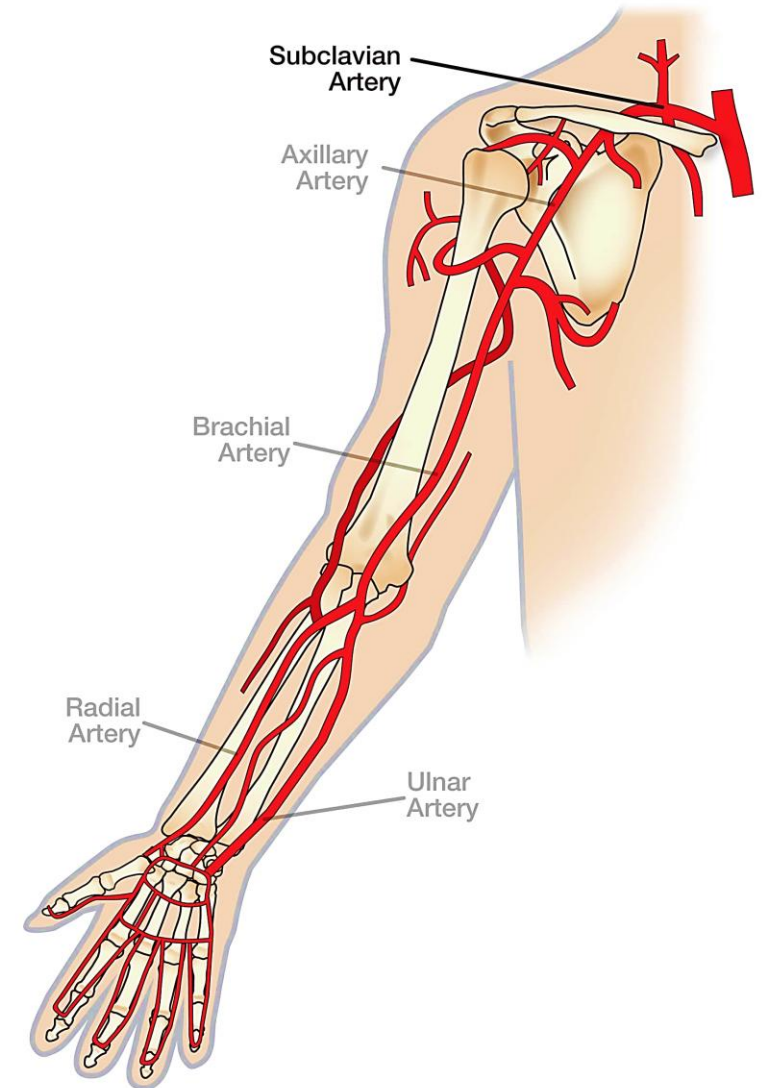
Subclavian artery

Axillary artery

Brachial artery

Radial artery

Ulnar artery

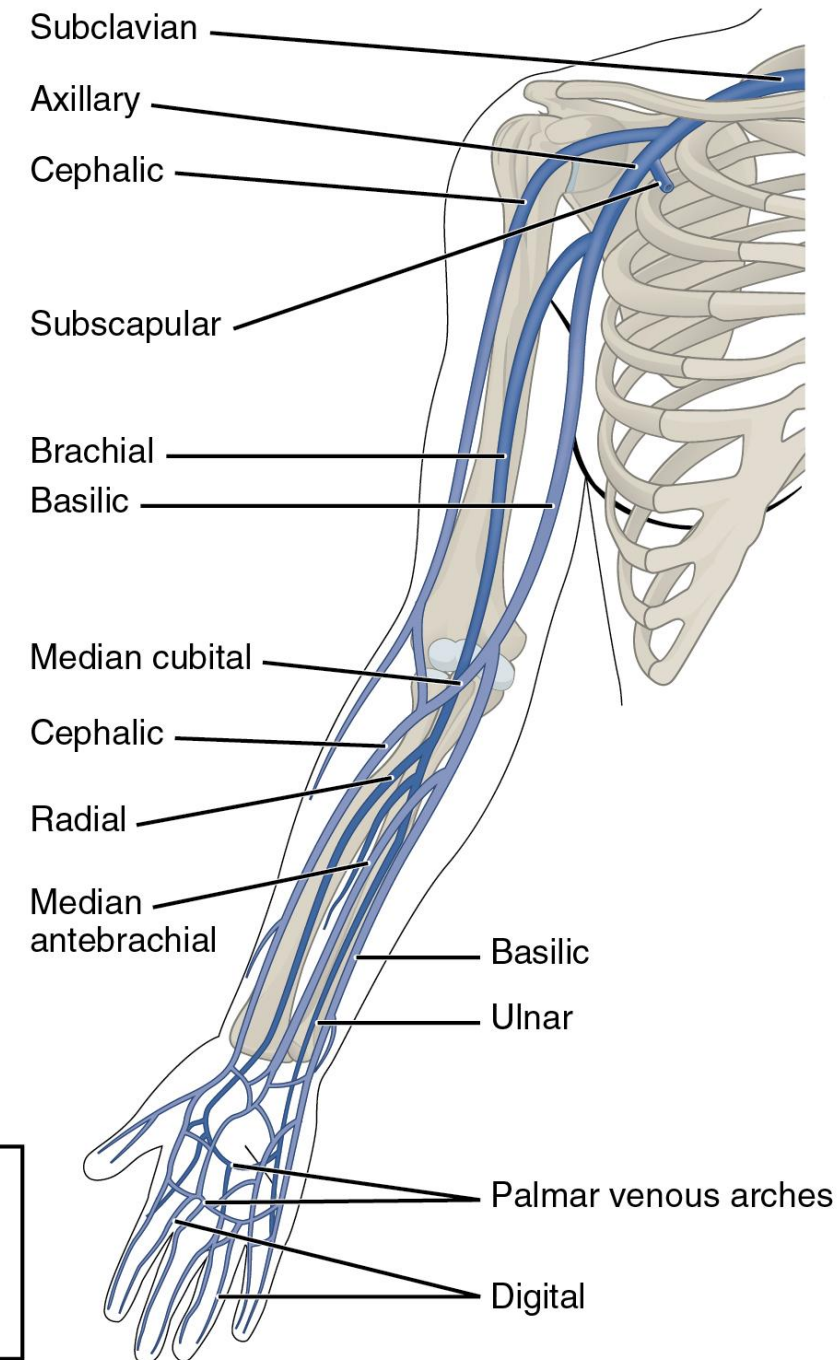
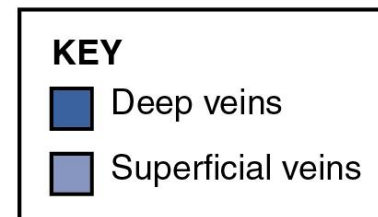


Venous drainage of the upper limb

- The major superficial veins of the upper limb are the **cephalic and basilic veins**.
 - a. **The basilic vein** originates from the **dorsal venous network of the hand**. It ascends the medial aspect of the upper limb. **terminates at axillary vein**.
 - b. **The cephalic vein** arises from the **dorsal venous network** of the hand. It ascends the antero-lateral aspect of the upper limb. **terminates by joining the axillary vein**.

Venous drainage of the upper limb

- **At the elbow**, the cephalic and basilic veins are connected by the **median cubital vein**.

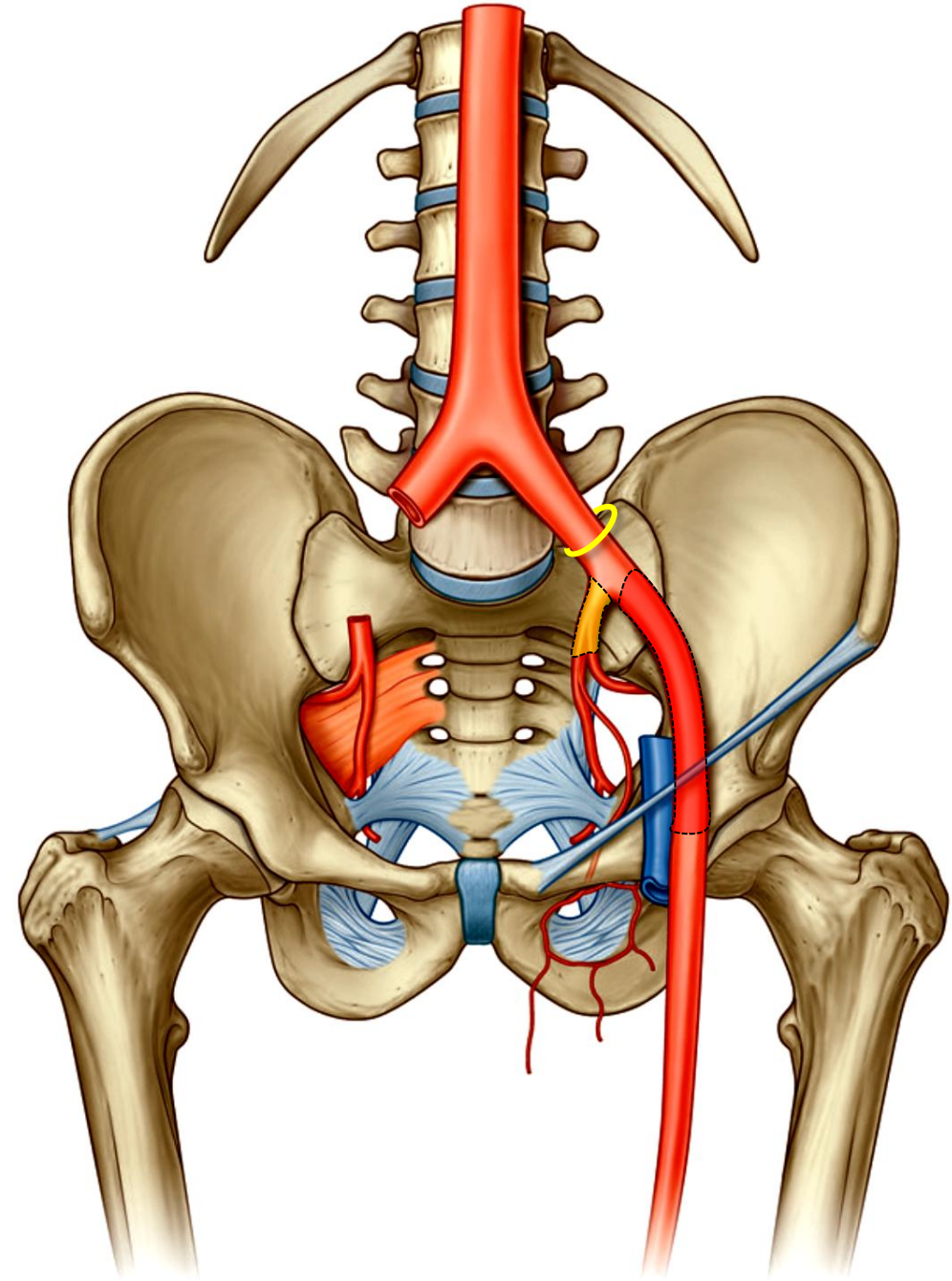


Venipuncture

- The venipuncture done in the **median cubital vein** because it is a superficial vein that is situated anteriorly at the elbow.
- It is commonly used due to its accessible and superficial position

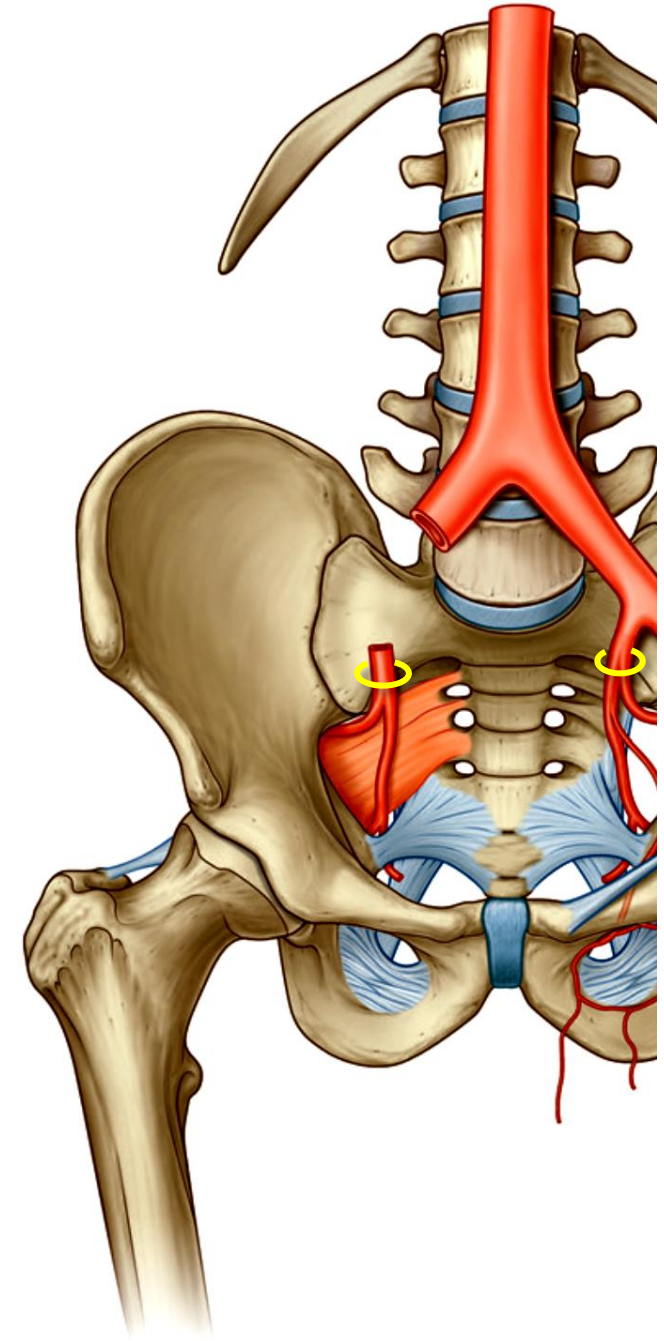
Common iliac arteries

- **At the level of L4**, the aorta terminates by **bifurcating** into:
 1. **the right and,**
 2. **left common iliac arteries** that supply the lower body.



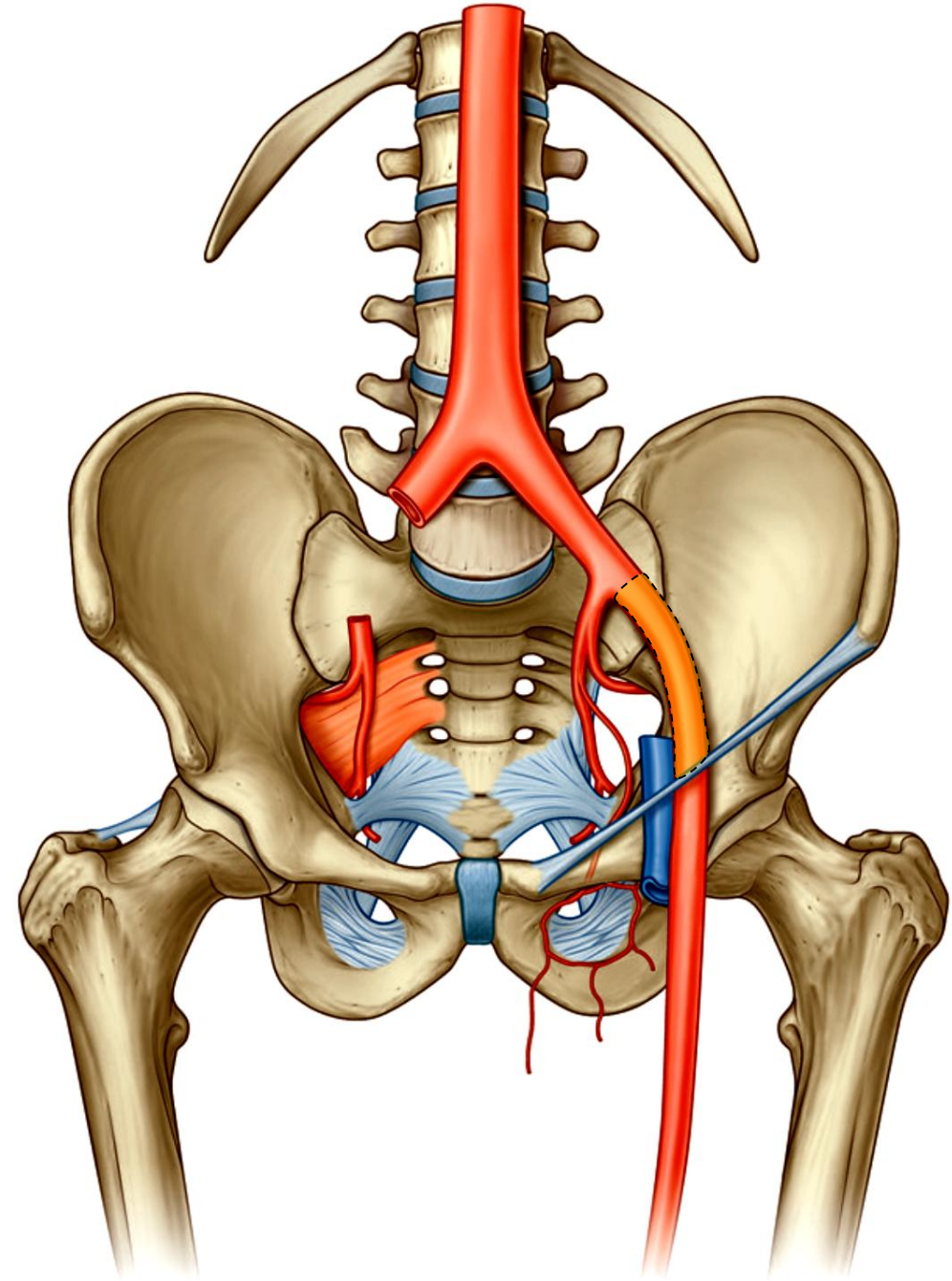
Internal iliac artery

- This artery passes down into the pelvic and its branches **supply the pelvic viscera, perineum, pelvic walls and buttock.**



External iliac artery

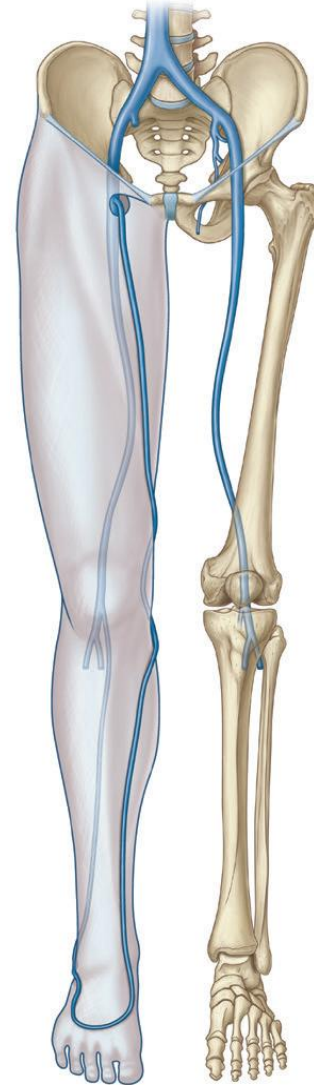
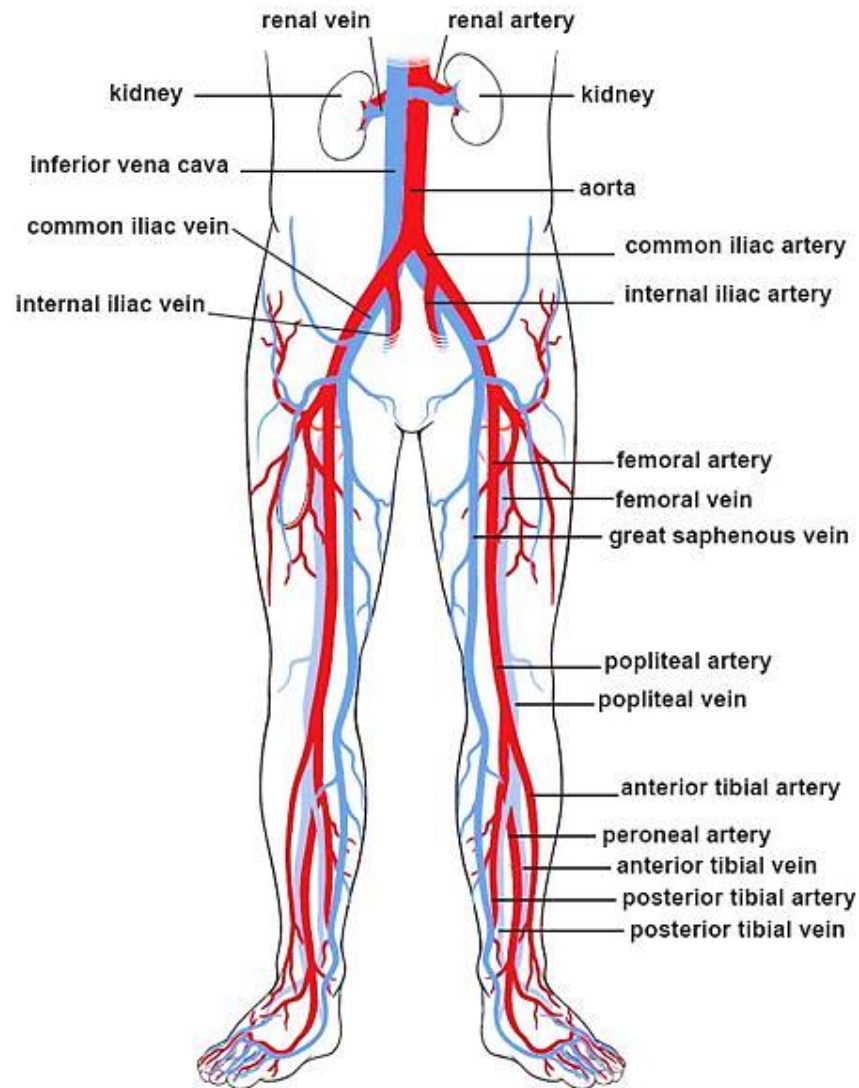
- The artery enters the thigh and becomes the **femoral artery**.



Common Iliac vein

- The common iliac vein is formed at the upper margin of the pubic symphysis by the **union of the external and internal iliac veins**.
- The external iliac vein is a **continuation of the femoral vein** (the major vessel draining the lower limb).
- The internal iliac vein is responsible the majority of **pelvic venous drainage**.

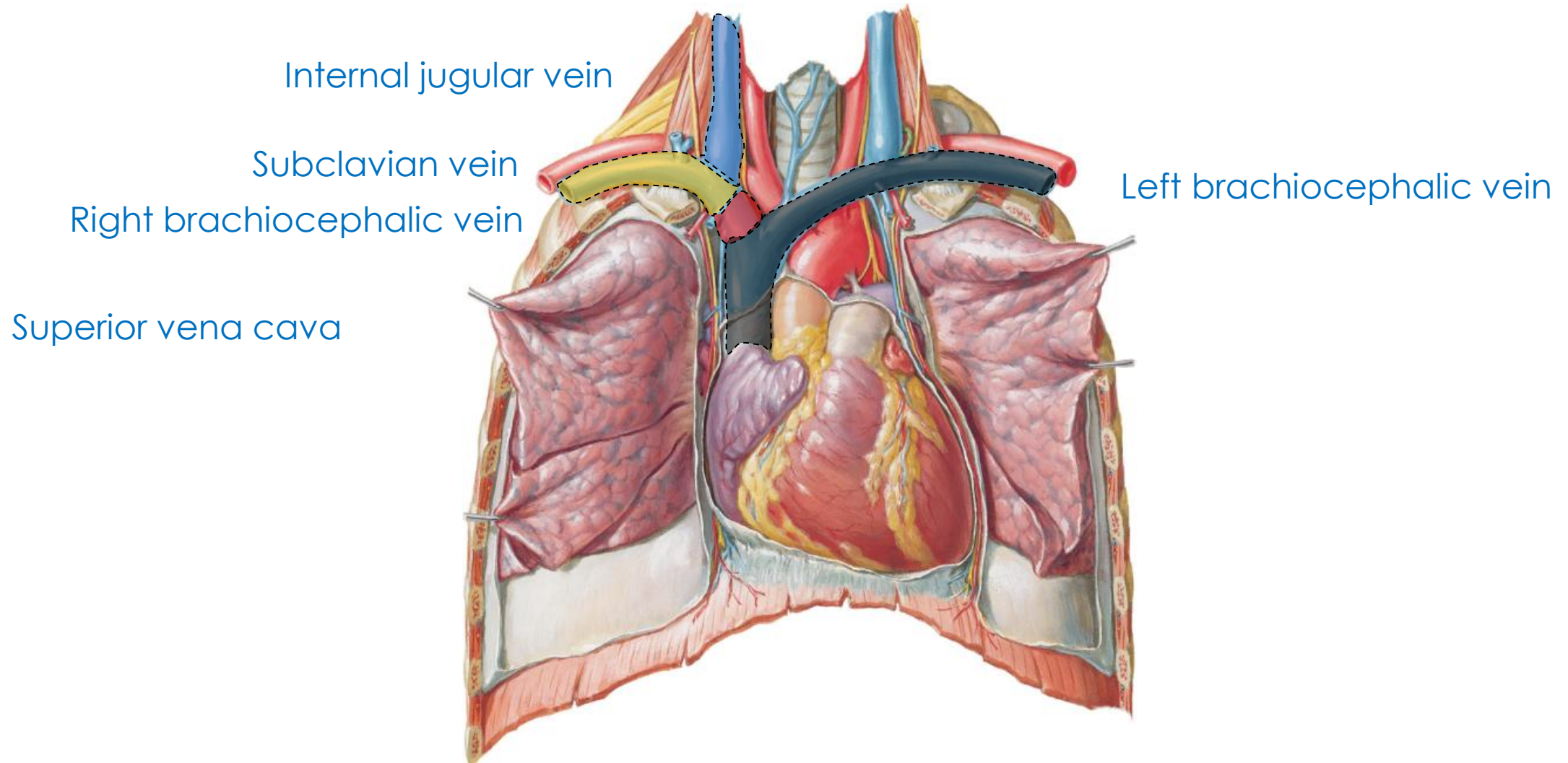
Venous drainage of the lower limb



Superior vena cava

- The superior vena cava receives deoxygenated blood from the upper body, and delivers it to the right atrium.
- It is formed by the Rt and Lt brachiocephalic veins joining together. It then travels through the thoracic region, draining into the right atrium.

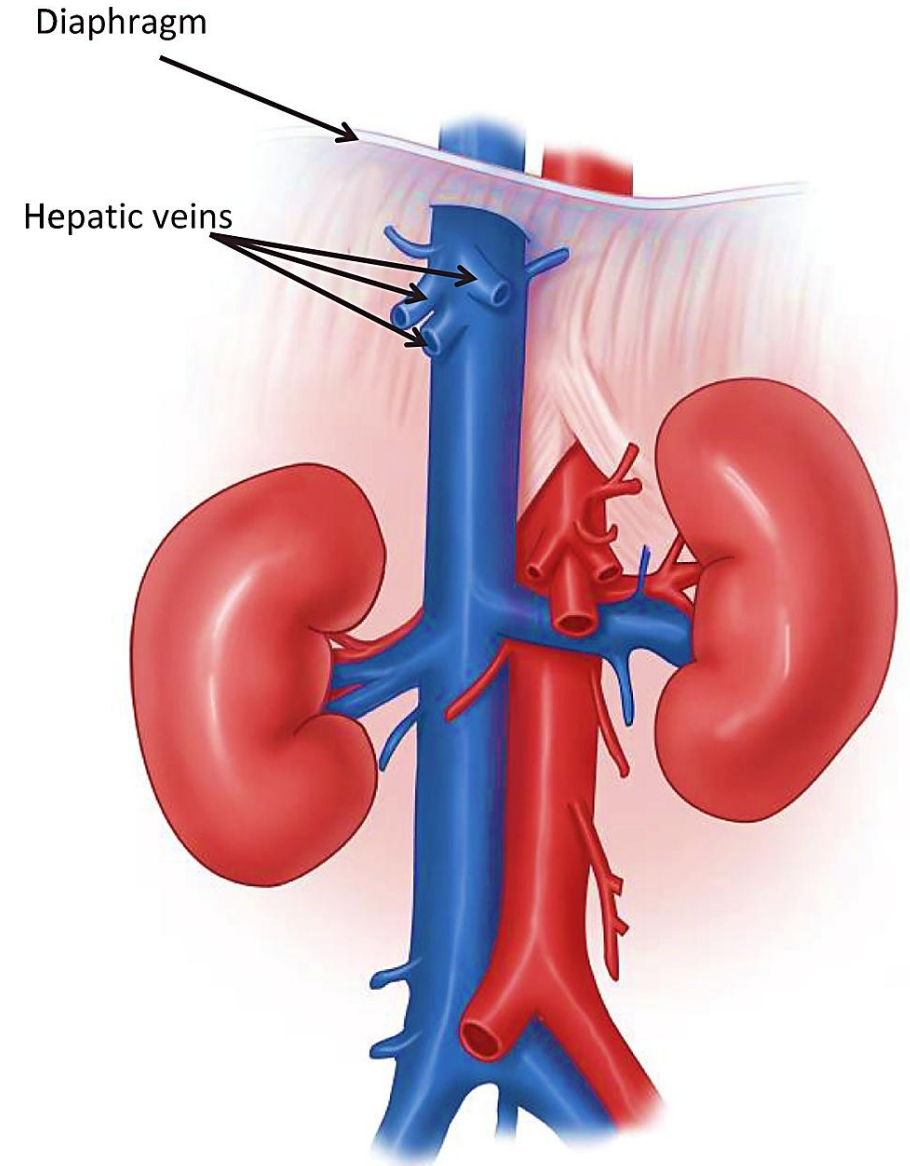
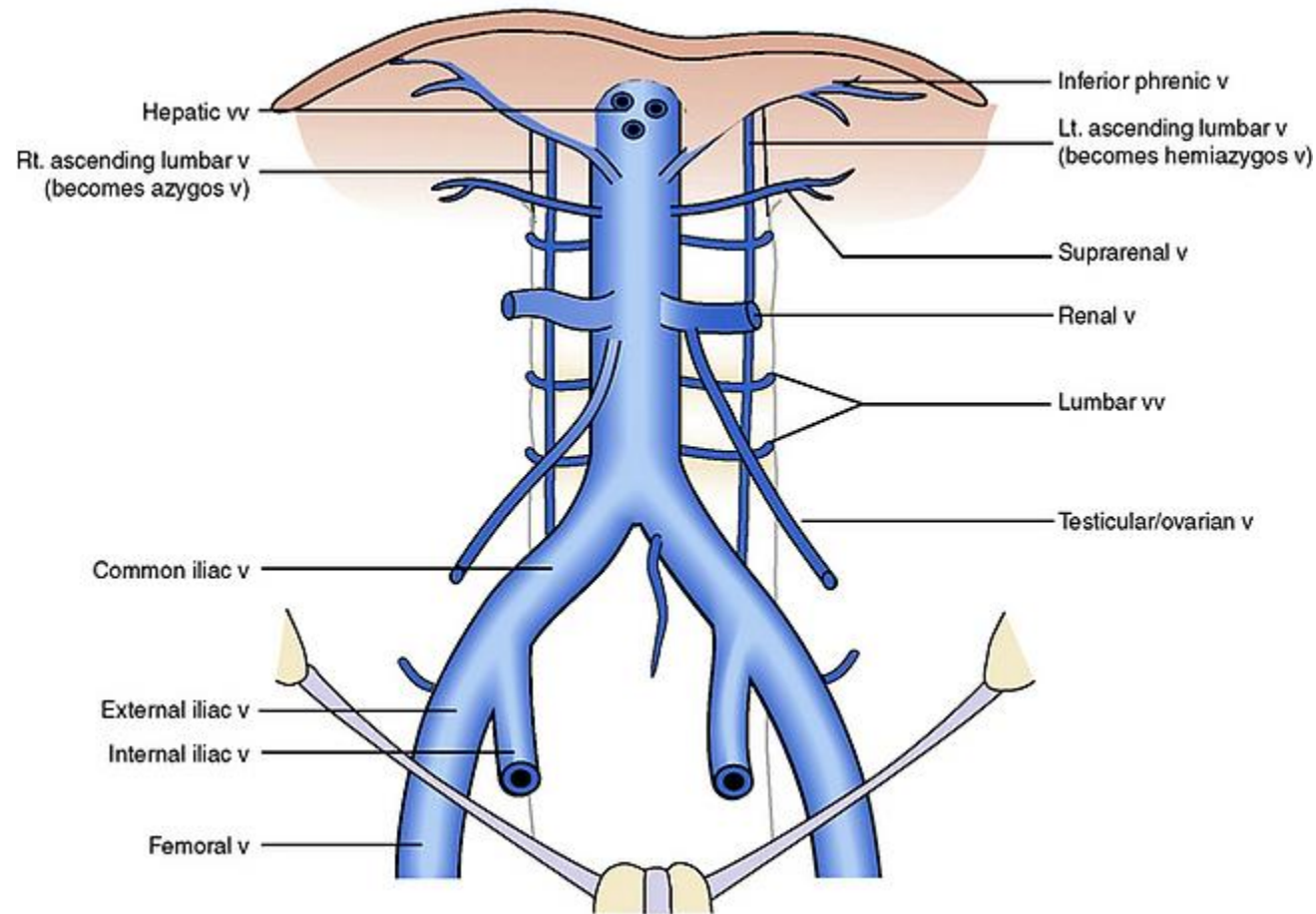
Superior vena cava



Inferior vena cava

- The inferior vena cava **receives deoxygenated blood from the lower body**, and delivers it to the **right atrium**.
- **The left and right common iliac veins** combine at **level L5** to become the **inferior vena cava**.
- The inferior vena cava then drains **into the right atrium**.

Inferior vena cava



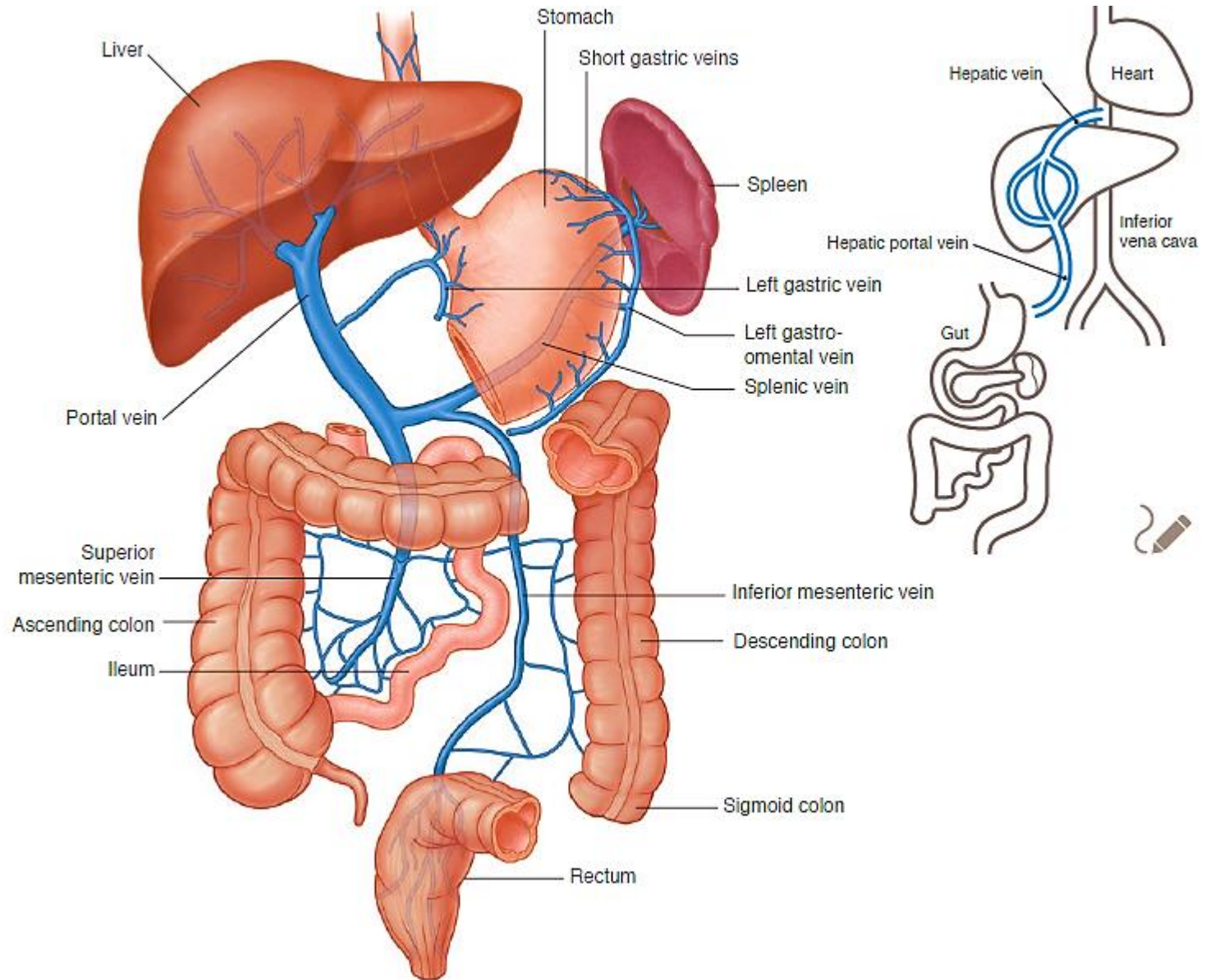
Venous drainage of abdomen

- There are two venous systems that drain abdominal structures – the **portal venous system** and the **systemic venous system**.
- **The portal system transports** venous blood to the liver for processing, whilst **the systemic venous system** returns blood to the right atrium of the heart.

Portal venous system

- **The portal system carries venous** blood (**rich in nutrients** that have been extracted from food) to the liver for processing.
- The major vessel of the portal system is **the portal vein**.
- It is the point of convergence for the **venous drainage of the spleen, pancreas, gallbladder and the abdominal part of the gastrointestinal tract**.

Portal venous system



Systemic venous system

- The systemic venous system transports deoxygenated blood **to the right atrium of the heart.**
- The major vessel in this system is **the inferior vena cava.**
- The vessel is formed by the **union of the common iliac veins at the L5 vertebral level.**